Local Projections vs VARs

Topics in Macroeconomics – Week 1

Lukas Boehnert

PhD elective, 2026/27

Bank of Portugal & Católica Lisbon

The Course

The Course

- **Five weeks, three parts:**
  - I) Empirical methods (Weeks 1–2): Local projections vs VARs, shift-share IV
  - II) International macro (Weeks 3–4): Financial crises, open-economy monetary policy
  - III) Macro-development (Week 5): Misallocation, spatial economics

The Course

- **Five weeks, three parts:**
  - I) Empirical methods (Weeks 1–2): Local projections vs VARs, shift-share IV
  - II) International macro (Weeks 3–4): Financial crises, open-economy monetary policy
  - III) Macro-development (Week 5): Misallocation, spatial economics

- **Each week:** $\sim$2 hours of lecture and coding, $\sim$1 hour of reading group

The Course

- **Five weeks, three parts:**
  - I) Empirical methods (Weeks 1–2): Local projections vs VARs, shift-share IV
  - II) International macro (Weeks 3–4): Financial crises, open-economy monetary policy
  - III) Macro-development (Week 5): Misallocation, spatial economics
- **Each week:** $\sim$2 hours of lecture and coding, $\sim$1 hour of reading group
- **Assessment:**
  - Class project (50%): apply one part of the course to your own research
  - Paper presentation (30%): one paper in the reading group, Weeks 2–4
  - Participation (20%): preparation, discussion, your turn as discussant

The Course

- **Five weeks, three parts:**
  - I) Empirical methods (Weeks 1–2): Local projections vs VARs, shift-share IV
  - II) International macro (Weeks 3–4): Financial crises, open-economy monetary policy
  - III) Macro-development (Week 5): Misallocation, spatial economics
- **Each week:** $\sim$2 hours of lecture and coding, $\sim$1 hour of reading group
- **Assessment:**
  - Class project (50%): apply one part of the course to your own research
  - Paper presentation (30%): one paper in the reading group, Weeks 2–4
  - Participation (20%): preparation, discussion, your turn as discussant

$\Rightarrow$ Goal: a project that could become the first draft of a thesis chapter

Today

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?
2. **Local projections:** Estimate the effect at each horizon directly

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?
2. **Local projections:** Estimate the effect at each horizon directly
3. **Structural VARs:** Model the whole system once, then iterate it forward

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?
2. **Local projections:** Estimate the effect at each horizon directly
3. **Structural VARs:** Model the whole system once, then iterate it forward
4. **Comparison:** Same example, both estimators: when do they agree, which should I use?

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?
2. **Local projections:** Estimate the effect at each horizon directly
3. **Structural VARs:** Model the whole system once, then iterate it forward
4. **Comparison:** Same example, both estimators: when do they agree, which should I use?
5. **Implementation:** How do I do it on real data? *(coding lab)*

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?
2. **Local projections:** Estimate the effect at each horizon directly
3. **Structural VARs:** Model the whole system once, then iterate it forward
4. **Comparison:** Same example, both estimators: when do they agree, which should I use?
5. **Implementation:** How do I do it on real data? *(coding lab)*
6. **Reading group:** Gertler & Karadi (2015) as a model discussion

Today

1. **Why impulse responses?** What do we want to measure, and where is it used?
2. **Local projections:** Estimate the effect at each horizon directly
3. **Structural VARs:** Model the whole system once, then iterate it forward
4. **Comparison:** Same example, both estimators: when do they agree, which should I use?
5. **Implementation:** How do I do it on real data? *(coding lab)*
6. **Reading group:** Gertler & Karadi (2015) as a model discussion

Material draws on Mavroeidis (Oxford Macroeconometrics), Plagborg-Møller (2026 LP lectures) and the Montiel Olea et al. (2025) primer.

Why impulse responses?

The object: a dynamic causal effect

The object: a dynamic causal effect

- The impulse response of variable i to structural shock j at horizon h:

$$IRF_{ij}(h) = \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 1\right] - \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 0\right]$$

The object: a dynamic causal effect

- The impulse response of variable i to structural shock j at horizon h:

$$IRF_{ij}(h) = \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 1\right] - \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 0\right]$$

- **Shock** $\varepsilon_{j,t}$: exogenous impulse, unrelated to other shocks at all leads and lags

The object: a dynamic causal effect

- The impulse response of variable i to structural shock j at horizon h:

$$IRF_{ij}(h) = \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 1\right] - \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 0\right]$$

- **Shock** $\varepsilon_{j,t}$: exogenous impulse, unrelated to other shocks at all leads and lags
- **Horizon** h: the whole path matters – timing, peak, persistence

The object: a dynamic causal effect

- The impulse response of variable i to structural shock j at horizon h:

$$IRF_{ij}(h) = \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 1\right] - \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 0\right]$$

- **Shock** $\varepsilon_{j,t}$: exogenous impulse, unrelated to other shocks at all leads and lags
- **Horizon** h: the whole path matters – timing, peak, persistence
- **Linearity**: maintained today $\rightarrow$ effect independent of size and sign of the shock

The object: a dynamic causal effect

- The impulse response of variable i to structural shock j at horizon h:

$$IRF_{ij}(h) = \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 1\right] - \mathbb{E}\left[y_{i,t+h} \mid \varepsilon_{j,t} = 0\right]$$

- **Shock** $\varepsilon_{j,t}$: exogenous impulse, unrelated to other shocks at all leads and lags
- **Horizon** h: the whole path matters – timing, peak, persistence
- **Linearity**: maintained today $\rightarrow$ effect independent of size and sign of the shock
- Three questions we ask of the data:
  1. Impulse response: what happens after a policy tightening, month by month?
  2. Variance decomposition: how important are policy shocks for output?
  3. Historical decomposition: which shocks drove the UK recession of the early 1990s?

[Stock & Watson (2018)]

Where impulse responses are used

Where impulse responses are used

- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]

Where impulse responses are used

- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]

Where impulse responses are used

- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]
- **Oil and commodities** [Kilian (2009), Känzig (2021)]

Where impulse responses are used

- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]
- **Oil and commodities** [Kilian (2009), Känzig (2021)]
- **Credit and crises** [Jordà, Schularick & Taylor (2013), Mian, Sufi & Verner (2017)]

Where impulse responses are used

- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]
- **Oil and commodities** [Kilian (2009), Känzig (2021)]
- **Credit and crises** [Jordà, Schularick & Taylor (2013), Mian, Sufi & Verner (2017)]
- **International spillovers** [Miranda-Agrippino & Rey (2020)]

Where impulse responses are used

- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]
- **Oil and commodities** [Kilian (2009), Känzig (2021)]
- **Credit and crises** [Jordà, Schularick & Taylor (2013), Mian, Sufi & Verner (2017)]
- **International spillovers** [Miranda-Agrippino & Rey (2020)]
- **Climate** [Bilal & Känzig (2025)]

Where impulse responses are used

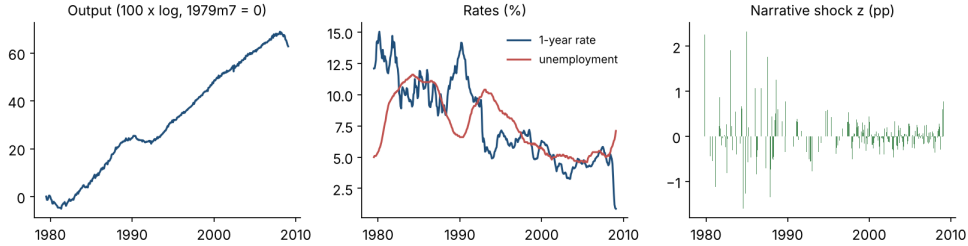
- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]
- **Oil and commodities** [Kilian (2009), Känzig (2021)]
- **Credit and crises** [Jordà, Schularick & Taylor (2013), Mian, Sufi & Verner (2017)]
- **International spillovers** [Miranda-Agrippino & Rey (2020)]
- **Climate** [Bilal & Känzig (2025)]
- **Policy institutions**: forecasting models, scenario analysis, historical decompositions

Where impulse responses are used

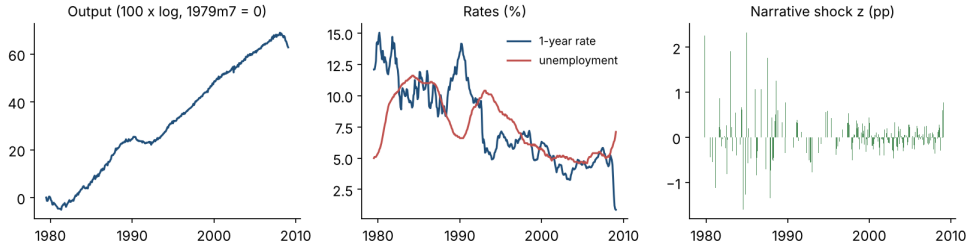
- **Monetary policy** [Christiano, Eichenbaum & Evans (1999), Gertler & Karadi (2015), Ramey (2016)]
- **Fiscal multipliers** [Blanchard & Perotti (2002), Ramey & Zubairy (2018)]
- **Oil and commodities** [Kilian (2009), Känzig (2021)]
- **Credit and crises** [Jordà, Schularick & Taylor (2013), Mian, Sufi & Verner (2017)]
- **International spillovers** [Miranda-Agrippino & Rey (2020)]
- **Climate** [Bilal & Känzig (2025)]
- **Policy institutions**: forecasting models, scenario analysis, historical decompositions

$\Rightarrow$ Weeks 3–5 lean on these tools too: crises, open-economy policy, regional shocks

Running example: UK monetary policy and output

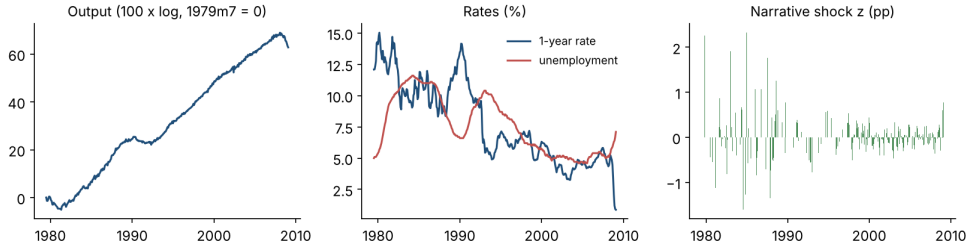


Running example: UK monetary policy and output



- Monthly, 1979m7–2009m2, $T = 356$ [data: Cesa-Bianchi, Thwaites & Vicondoa (2020)]
  - y: output ($100 \times$ log monthly GDP), u: unemployment rate, i: 1-year gilt yield
  - z: narrative monetary policy shock [Cloyne & Hürtgen (2016)]

Running example: UK monetary policy and output



- Monthly, 1979m7–2009m2, $T = 356$ [data: Cesa-Bianchi, Thwaites & Vicondoa (2020)]
  - y: output ($100 \times$ log monthly GDP), u: unemployment rate, i: 1-year gilt yield
  - z: narrative monetary policy shock [Cloyne & Hürtgen (2016)]
- **Question:** what does a 1pp higher 1-year rate do to output, month by month?

Why correlations are not enough

Why correlations are not enough

- UK 1979–2009: regress output on the 1-year rate $\rightarrow$ -6% per pp, at every horizon. Is that what policy does?

Why correlations are not enough

- UK 1979–2009: regress output on the 1-year rate $\rightarrow$ -6% per pp, at every horizon. Is that what policy does?
- Two equations generate the same comovement:

$$\text{Transmission:} \quad y_t = \beta\, i_t + \omega_y \eta_{y,t}$$

$$\text{Policy rule:} \quad i_t = \gamma\, y_t + \omega_i \eta_{i,t}$$

Why correlations are not enough

- UK 1979–2009: regress output on the 1-year rate $\rightarrow$ -6% per pp, at every horizon. Is that what policy does?
- Two equations generate the same comovement:

$$\text{Transmission:} \quad y_t = \beta\, i_t + \omega_y \eta_{y,t}$$

$$\text{Policy rule:} \quad i_t = \gamma\, y_t + \omega_i \eta_{i,t}$$

- The correlation mixes transmission, policy reaction and common trends

Why correlations are not enough

- UK 1979–2009: regress output on the 1-year rate $\rightarrow$ -6% per pp, at every horizon. Is that what policy does?
- Two equations generate the same comovement:

$$\text{Transmission:} \quad y_t = \beta\, i_t + \omega_y \eta_{y,t}$$

$$\text{Policy rule:} \quad i_t = \gamma\, y_t + \omega_i \eta_{i,t}$$

- The correlation mixes transmission, policy reaction and common trends
- Random rate experiments are infeasible $\Rightarrow$ need either
  - a natural experiment (instrument), or
  - assumptions (timing, signs, ...)

[Mavroeidis, Oxford Macroeconometrics, lecture 4]

Two separate problems

Two separate problems

- **Identification** = which population object equals the causal effect? *(economics)*

Two separate problems

- **Identification** = which population object equals the causal effect? *(economics)*
  - Timing (recursive ordering), external instruments, sign restrictions, ...

Two separate problems

- **Identification** = which population object equals the causal effect? *(economics)*
  - Timing (recursive ordering), external instruments, sign restrictions, ...
- **Estimation** = how do we compute it from a finite sample? *(statistics)*

Two separate problems

- **Identification** = which population object equals the causal effect? *(economics)*
  - Timing (recursive ordering), external instruments, sign restrictions, ...
- **Estimation** = how do we compute it from a finite sample? *(statistics)*
  - Iterate a VAR forward, project directly at each horizon (LP), shrink (BVAR, smooth LP)

Two separate problems

- **Identification** = which population object equals the causal effect? *(economics)*
  - Timing (recursive ordering), external instruments, sign restrictions, ...
- **Estimation** = how do we compute it from a finite sample? *(statistics)*
  - Iterate a VAR forward, project directly at each horizon (LP), shrink (BVAR, smooth LP)

$\Rightarrow$ Any identification scheme works with either estimator

Two separate problems

- **Identification** = which population object equals the causal effect? *(economics)*
  - Timing (recursive ordering), external instruments, sign restrictions, ...
- **Estimation** = how do we compute it from a finite sample? *(statistics)*
  - Iterate a VAR forward, project directly at each horizon (LP), shrink (BVAR, smooth LP)

$\Rightarrow$ Any identification scheme works with either estimator

$\Rightarrow$ The LP vs VAR debate is about estimation, not identification [Plagborg-Møller & Wolf (2021)]

One example, three passes

One example, three passes

- **LPs** (theory, then the example):
  L1 No controls L2 Lags and date-t controls L3 Instrument L4 Lags with an instrument
  L5 Standard errors L6 Variance decomposition

One example, three passes

- **LPs** (theory, then the example):
  - L1 No controls
  - L2 Lags and date-t controls
  - L3 Instrument
  - L4 Lags with an instrument
  - L5 Standard errors
  - L6 Variance decomposition
- **VARs** (theory, then the example):
  - V1 Recursive VAR
  - V2 Ordering
  - V3 Lag length
  - V4 Instrument
  - V5 Variance decomposition

One example, three passes

- **LPs** (theory, then the example):
  L1 No controls L2 Lags and date-t controls L3 Instrument L4 Lags with an instrument
  L5 Standard errors L6 Variance decomposition
- **VARs** (theory, then the example):
  V1 Recursive VAR V2 Ordering V3 Lag length V4 Instrument V5 Variance decomposition
- **Comparison** (same shock, both estimators):
  C1 Horizon by horizon C2 One instrument, three estimators C3 Bias vs variance C4 Variance decompositions

One example, three passes

- **LPs** (theory, then the example):
  L1 No controls L2 Lags and date-t controls L3 Instrument L4 Lags with an instrument
  L5 Standard errors L6 Variance decomposition
- **VARs** (theory, then the example):
  V1 Recursive VAR V2 Ordering V3 Lag length V4 Instrument V5 Variance decomposition
- **Comparison** (same shock, both estimators):
  C1 Horizon by horizon C2 One instrument, three estimators C3 Bias vs variance C4 Variance decompositions

$\Rightarrow$ Every number and figure comes from `week1_simple.py` (one cell per step) and `week1_fast.py`

Local projections

LPs: one regression per horizon, no iterating

LPs: one regression per horizon, no iterating

- AR(1) iterated h steps forward:

$$y_{t+h} = \rho^h y_t + \underbrace{\varepsilon_{t+h} + \rho\,\varepsilon_{t+h-1} + \cdots + \rho^{h-1}\varepsilon_{t+1}}_{\text{forecast error}}$$

LPs: one regression per horizon, no iterating

- AR(1) iterated h steps forward:

$$y_{t+h} = \rho^h y_t + \underbrace{\varepsilon_{t+h} + \rho\,\varepsilon_{t+h-1} + \cdots + \rho^{h-1}\varepsilon_{t+1}}_{\text{forecast error}}$$

- **LP:** project y_{t+h} on y_t directly, a separate regression for each $h \to$ the slope estimates ρ^h from the data at lag h

LPs: one regression per horizon, no iterating

- AR(1) iterated h steps forward:

$$y_{t+h} = \rho^h y_t + \underbrace{\varepsilon_{t+h} + \rho\,\varepsilon_{t+h-1} + \cdots + \rho^{h-1}\varepsilon_{t+1}}_{\text{forecast error}}$$

- **LP:** project y_{t+h} on y_t directly, a separate regression for each $h \to$ the slope estimates ρ^h from the data at lag h
- **VAR:** estimate $\hat{\rho}$ once, report $\hat{\rho}^h \to$ iterates the one-step model

LPs: one regression per horizon, no iterating

- AR(1) iterated h steps forward:

$$y_{t+h} = \rho^h y_t + \underbrace{\varepsilon_{t+h} + \rho\,\varepsilon_{t+h-1} + \cdots + \rho^{h-1}\varepsilon_{t+1}}_{\text{forecast error}}$$

- **LP:** project y_{t+h} on y_t directly, a separate regression for each $h \to$ the slope estimates ρ^h from the data at lag h
- **VAR:** estimate $\hat{\rho}$ once, report $\hat{\rho}^h \to$ iterates the one-step model
- Why applied people like it: just OLS; nonlinearities, panels, IV are one line each

LPs: one regression per horizon, no iterating

- AR(1) iterated h steps forward:

$$y_{t+h} = \rho^h y_t + \underbrace{\varepsilon_{t+h} + \rho\,\varepsilon_{t+h-1} + \cdots + \rho^{h-1}\varepsilon_{t+1}}_{\text{forecast error}}$$

- **LP:** project y_{t+h} on y_t directly, a separate regression for each $h \to$ the slope estimates ρ^h from the data at lag h
- **VAR:** estimate $\hat{\rho}$ once, report $\hat{\rho}^h \to$ iterates the one-step model
- Why applied people like it: just OLS; nonlinearities, panels, IV are one line each
- What it is not: a model of the system, or free identification

[Jordà (2005)]

The LP regression

The LP regression

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, H$$

The LP regression

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, H$$

- y_{t+h}: outcome — UK: output y_{t+h}
- x_t: impulse variable — 1-year rate i_t
- r_t: contemporaneous controls (not moved by the shock within the period) — y_t, u_t
- $w_{t-1}, \ldots, w_{t-p}$: lagged controls; always include lags of y and x — 12 lags of y, u, i
- θ_h: the impulse response at horizon h

The LP regression

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, H$$

- y_{t+h}: outcome — UK: output y_{t+h}
- x_t: impulse variable — 1-year rate i_t
- r_t: contemporaneous controls (not moved by the shock within the period) — y_t, u_t
- $w_{t-1}, \ldots, w_{t-p}$: lagged controls; always include lags of y and x — 12 lags of y, u, i
- θ_h: the impulse response at horizon h

$\Rightarrow$ y and x come from the question; the controls are the identifying assumption

[Plagborg-Møller (2026), lecture 1]

What is an LP estimating? The implicit shock

What is an LP estimating? The implicit shock

- Frisch-Waugh: the LP coefficient θ_h on x_t, with $\tilde{x}_t$ = residual of x_t on r_t and the lags, is

$$\theta_h = \frac{\text{Cov}(y_{t+h}, \tilde{x}_t)}{\text{Var}(\tilde{x}_t)}$$

What is an LP estimating? The implicit shock

- Frisch-Waugh: the LP coefficient θ_h on x_t, with $\tilde{x}_t$ = residual of x_t on r_t and the lags, is

$$\theta_h = \frac{\text{Cov}(y_{t+h}, \tilde{x}_t)}{\text{Var}(\tilde{x}_t)}$$

$\Rightarrow$ LP measures the response to a one-unit move in $\tilde{x}_t$: the controls define the shock

What is an LP estimating? The implicit shock

- Frisch-Waugh: the LP coefficient θ_h on x_t, with $\tilde{x}_t$ = residual of x_t on r_t and the lags, is

$$\theta_h = \frac{\text{Cov}(y_{t+h}, \tilde{x}_t)}{\text{Var}(\tilde{x}_t)}$$

$\Rightarrow$ LP measures the response to a one-unit move in $\tilde{x}_t$: the controls define the shock

- A sensible $\tilde{x}_t$ should be
  1. serially uncorrelated (else add lags)
  2. unpredictable by other lagged series (else add controls, rethink reverse causality)
  3. a shock with a model counterpart (what would a "wage shock" be?)

What is an LP estimating? The implicit shock

- Frisch-Waugh: the LP coefficient θ_h on x_t, with $\tilde{x}_t$ = residual of x_t on r_t and the lags, is

$$\theta_h = \frac{\text{Cov}(y_{t+h}, \tilde{x}_t)}{\text{Var}(\tilde{x}_t)}$$

$\Rightarrow$ LP measures the response to a one-unit move in $\tilde{x}_t$: the controls define the shock

- A sensible $\tilde{x}_t$ should be
  1. serially uncorrelated (else add lags)
  2. unpredictable by other lagged series (else add controls, rethink reverse causality)
  3. a shock with a model counterpart (what would a "wage shock" be?)

- **In practice:** run one regression with all controls (second-step SEs are wrong), but plot $\tilde{x}_t$ and report its s.d.

  Climate example: sd $\approx$ 0.1°C, so "+1°C" is a huge shock [Bilal & Känzig (2025)]

Identification methods

Identification methods

- Goal: make $\tilde{x}_t$ a shock. Three ways to get there:

Identification methods

- Goal: make $\tilde{x}_t$ a shock. Three ways to get there:

1) **Timing restrictions:** control for everything policy reacts to (lags, date-t controls)
   - UK: i_t on y_t, u_t and lags (L1, L2)
   - Often implausible: we cannot control for every endogenous factor (forecasts, inflation, sterling, ...)

Identification methods

- Goal: make $\tilde{x}_t$ a shock. Three ways to get there:

1) **Timing restrictions:** control for everything policy reacts to (lags, date-t controls)
   - UK: i_t on y_t, u_t and lags (L1, L2)
   - Often implausible: we cannot control for every endogenous factor (forecasts, inflation, sterling, ...)
2) **Observed shocks:** measure the shock directly and use it as x_t
   - Oil supply news [Känzig (2021)], narrative tax/policy shocks [Romer & Romer (2004, 2010)]

Identification methods

- Goal: make $\tilde{x}_t$ a shock. Three ways to get there:

1) **Timing restrictions:** control for everything policy reacts to (lags, date-t controls)
   - UK: i_t on y_t, u_t and lags (L1, L2)
   - Often implausible: we cannot control for every endogenous factor (forecasts, inflation, sterling, ...)

2) **Observed shocks:** measure the shock directly and use it as x_t
   - Oil supply news [Känzig (2021)], narrative tax/policy shocks [Romer & Romer (2004, 2010)]

3) **Instruments (LP-IV):** the shock series is a noisy proxy; use it to instrument x_t
   - UK high-frequency surprises [Cesa-Bianchi, Thwaites & Vicondoa (2020)], narrative shock [Cloyne & Hürtgen (2016)] (L3, L4)

Identification methods

- Goal: make $\tilde{x}_t$ a shock. Three ways to get there:

1) **Timing restrictions:** control for everything policy reacts to (lags, date-t controls)
   - UK: i_t on y_t, u_t and lags (L1, L2)
   - Often implausible: we cannot control for every endogenous factor (forecasts, inflation, sterling, ...)

2) **Observed shocks:** measure the shock directly and use it as x_t
   - Oil supply news [Känzig (2021)], narrative tax/policy shocks [Romer & Romer (2004, 2010)]

3) **Instruments (LP-IV):** the shock series is a noisy proxy; use it to instrument x_t
   - UK high-frequency surprises [Cesa-Bianchi, Thwaites & Vicondoa (2020)], narrative shock [Cloyne & Hürtgen (2016)] (L3, L4)

$\Rightarrow$ Let's start with 1): timing restrictions and adding lags

Identification by timing restrictions

Identification by timing restrictions

- **Assumptions:**
  - Contemporaneous controls r_t do not respond to the shock within the period
  - x_t responds within the period only to r_t and to the shock of interest

Identification by timing restrictions

- **Assumptions:**
  - Contemporaneous controls r_t do not respond to the shock within the period
  - x_t responds within the period only to r_t and to the shock of interest
- If they fail: r_t "over-controls" or $\tilde{x}_t$ mixes several shocks

Identification by timing restrictions

- **Assumptions:**
  - Contemporaneous controls r_t do not respond to the shock within the period
  - x_t responds within the period only to r_t and to the shock of interest
- If they fail: r_t "over-controls" or $\tilde{x}_t$ mixes several shocks
- **Running example:** y_{t+h} on i_t, controlling for y_t, u_t and lags of y, u, i
  $\rightarrow$ assumption: the rate does not move output or unemployment within the month

Identification by timing restrictions

- **Assumptions:**
  - Contemporaneous controls r_t do not respond to the shock within the period
  - x_t responds within the period only to r_t and to the shock of interest
- If they fail: r_t "over-controls" or $\tilde{x}_t$ mixes several shocks
- **Running example:** y_{t+h} on i_t, controlling for y_t, u_t and lags of y, u, i
  $\rightarrow$ assumption: the rate does not move output or unemployment within the month

$\Rightarrow$ Same logic as a recursive (Cholesky) ordering: we meet it again as the recursive VAR in V1

Choosing controls and lags: a recipe

Choosing controls and lags: a recipe

1. **Controls from economics:** list channels of reverse causality ($y \to x$); state timing assumptions

Choosing controls and lags: a recipe

1. **Controls from economics:** list channels of reverse causality ($y \to x$); state timing assumptions
2. **Always include lags of y and x.** Drop a control only if insignificant for both y_{t+h} and x_t

Choosing controls and lags: a recipe

1. **Controls from economics:** list channels of reverse causality ($y \to x$); state timing assumptions
2. **Always include lags of y and x.** Drop a control only if insignificant for both y_{t+h} and x_t
3. **Lag length:** fit an auxiliary VAR in all series, choose p by AIC, discard the VAR, use p lags in every LP

Choosing controls and lags: a recipe

1. **Controls from economics:** list channels of reverse causality ($y \to x$); state timing assumptions
2. **Always include lags of y and x.** Drop a control only if insignificant for both y_{t+h} and x_t
3. **Lag length:** fit an auxiliary VAR in all series, choose p by AIC, discard the VAR, use p lags in every LP
4. **If in doubt, more lags**: they cost little and soak up residual variance

Choosing controls and lags: a recipe

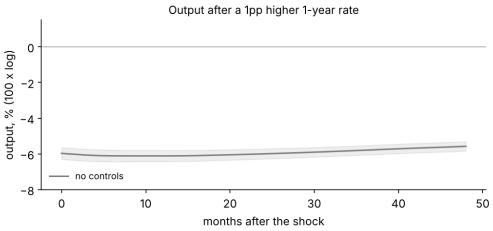
1. **Controls from economics:** list channels of reverse causality ($y \to x$); state timing assumptions
2. **Always include lags of y and x.** Drop a control only if insignificant for both y_{t+h} and x_t
3. **Lag length:** fit an auxiliary VAR in all series, choose p by AIC, discard the VAR, use p lags in every LP
4. **If in doubt, more lags**: they cost little and soak up residual variance

- **Observed shock?** [e.g. OPEC news surprises, Känzig (2021)]
  - Controls not needed for consistency, but add lags of y, x to shrink SEs
  - If adding lags moves the estimate a lot $\to$ your "shock" is predictable

[Plagborg-Møller (2026), lecture 1]

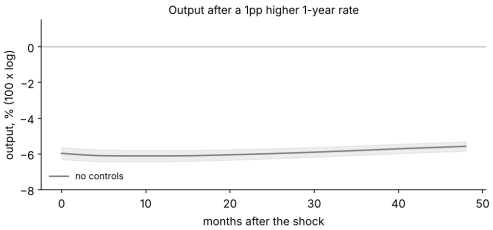
L1: no controls

$$y_{t+h} = \mu_h + \theta_h \, i_t + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, 48$$



L1: no controls

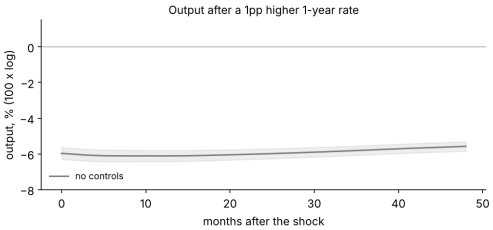
$$y_{t+h} = \mu_h + \theta_h \, i_t + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, 48$$



- −6.0% on impact, −6.1% at 12 months, −6.0% at 24

L1: no controls

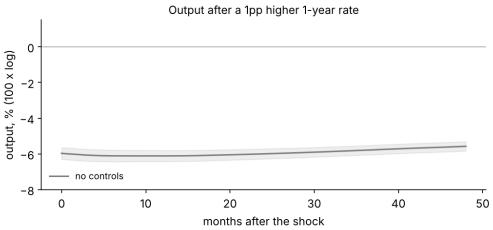
$$y_{t+h} = \mu_h + \theta_h \, i_t + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, 48$$



- -6.0% on impact, -6.1% at 12 months, -6.0% at 24
- A flat line from $h = 0$: not transmission, but a common trend (output up, rates down over 1979–2009)

L1: no controls

$$y_{t+h} = \mu_h + \theta_h \, i_t + \xi_{h,t+h}, \qquad h = 0, 1, \ldots, 48$$

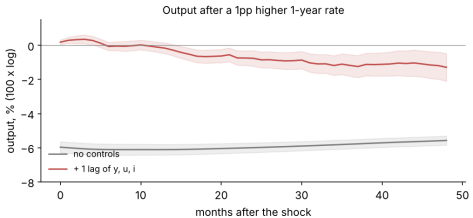


- -6.0% on impact, -6.1% at 12 months, -6.0% at 24
- A flat line from $h = 0$: not transmission, but a common trend (output up, rates down over 1979–2009)

$\Rightarrow$ $\tilde{x}_t = i_t$: the "shock" is the level of the rate, which nobody believes is random

L2: add lags, then date-t controls

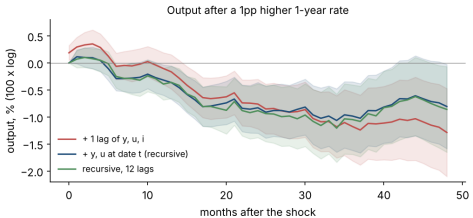
$$y_{t+h} = \mu_h + \theta_h\, i_t + \delta_h' x_{t-1} + \xi_{h,t+h}, \qquad x_t = (y_t, u_t, i_t)'$$



- **+ 1 lag of** y, u, i**:** trend gone, but output rises on impact ($+0.19$): policy reacting to output

L2: add lags, then date-t controls

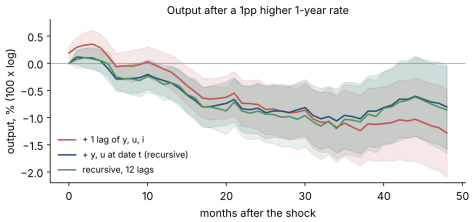
$$y_{t+h} = \mu_h + \theta_h\, i_t + \gamma_{1h}\, y_t + \gamma_{2h}\, u_t + \delta_h' x_{t-1} + \xi_{h,t+h}$$



- **+ 1 lag of** y, u, i**:** trend gone, but output rises on impact ($+0.19$): policy reacting to output
- **+** y_t, u_t **(recursive):** impact $= 0$ by assumption; -0.32% at 12 months, -0.83% at 24

L2: add lags, then date-t controls

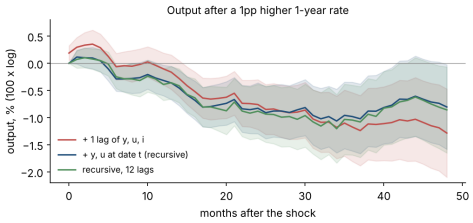
$$y_{t+h} = \mu_h + \theta_h\, i_t + \gamma_{1h}\, y_t + \gamma_{2h}\, u_t + \sum_{\ell=1}^{12} \delta'_{h,\ell} x_{t-\ell} + \xi_{h,t+h}$$



- **+ 1 lag of** y, u, i**:** trend gone, but output rises on impact ($+0.19$): policy reacting to output
- **+** y_t, u_t **(recursive):** impact $= 0$ by assumption; -0.32% at 12 months, -0.83% at 24
- **12 lags:** -0.39% and -0.88% $\rightarrow$ more lags barely move it

L2: add lags, then date-t controls

$$y_{t+h} = \mu_h + \theta_h\, i_t + \gamma_{1h}\, y_t + \gamma_{2h}\, u_t + \sum_{\ell=1}^{12} \delta'_{h,\ell} x_{t-\ell} + \xi_{h,t+h}$$



- **+ 1 lag of** y, u, i**:** trend gone, but output rises on impact ($+0.19$): policy reacting to output
- **+** y_t, u_t **(recursive):** impact $= 0$ by assumption; -0.32% at 12 months, -0.83% at 24
- **12 lags:** -0.39% and -0.88% $\rightarrow$ more lags barely move it

$\Rightarrow$ Lags remove what is predictable; date-t controls add a timing assumption (V1)

Standard errors: use lag augmentation

Standard errors: use lag augmentation

- LP error is a multi-step forecast error $\rightarrow$ serially correlated:

$$y_{t+2} = \rho^2 y_t + \underbrace{\varepsilon_{t+2} + \rho\,\varepsilon_{t+1}}_{\xi_{2,t+2}}$$

Standard errors: use lag augmentation

- LP error is a multi-step forecast error $\rightarrow$ serially correlated:

$$y_{t+2} = \rho^2 y_t + \underbrace{\varepsilon_{t+2} + \rho\,\varepsilon_{t+1}}_{\xi_{2,t+2}}$$

- **Old default:** Newey-West HAC with a bandwidth
  $\rightarrow$ often severely too small in samples like ours [Lazarus, Lewis, Stock & Watson (2018)]

Standard errors: use lag augmentation

- LP error is a multi-step forecast error → serially correlated:

$$y_{t+2} = \rho^2 y_t + \underbrace{\varepsilon_{t+2} + \rho\,\varepsilon_{t+1}}_{\xi_{2,t+2}}$$

- **Old default:** Newey-West HAC with a bandwidth
  → often severely too small in samples like ours [Lazarus, Lewis, Stock & Watson (2018)]
- **Better:** lag-augmented LP [Montiel Olea & Plagborg-Møller (2021)]
  - Control for lags of the data (+1 extra) → plain heteroskedasticity-robust SEs are valid
  - No bandwidth; robust to persistent data at moderate horizons
  - Needs enough lags (AIC) and a shock not predictable by future shocks; check: [Xu (2023)]

Standard errors: use lag augmentation

- LP error is a multi-step forecast error → serially correlated:

$$y_{t+2} = \rho^2 y_t + \underbrace{\varepsilon_{t+2} + \rho\,\varepsilon_{t+1}}_{\xi_{2,t+2}}$$

- **Old default:** Newey-West HAC with a bandwidth
  → often severely too small in samples like ours [Lazarus, Lewis, Stock & Watson (2018)]
- **Better:** lag-augmented LP [Montiel Olea & Plagborg-Møller (2021)]
  - Control for lags of the data (+1 extra) → plain heteroskedasticity-robust SEs are valid
  - No bandwidth; robust to persistent data at moderate horizons
  - Needs enough lags (AIC) and a shock not predictable by future shocks; check: [Xu (2023)]
- Pointwise bands: $\hat{\theta}_h \pm 1 \cdot SE$ (68%), $\pm 1.645 \cdot SE$ (90%); whole-path claims need a sup-t band [Montiel Olea & Plagborg-Møller (2019)]

LP-IV: when you only have a noisy proxy

LP-IV: when you only have a noisy proxy

- Proxy with measurement error: $z_t = \alpha + \beta\,\varepsilon_t + v_t \rightarrow$ LP of y on z attenuated

LP-IV: when you only have a noisy proxy

- Proxy with measurement error: $z_t = \alpha + \beta\,\varepsilon_t + v_t \rightarrow$ LP of y on z attenuated
- **Unit-effect normalization:** attenuation cancels in the ratio

$$\theta_h = \frac{\theta_{z\to y,h}}{\theta_{z\to s,0}} \quad = \quad \text{2SLS of } y_{t+h} \text{ on } s_t, \text{ instrumented by } z_t$$

LP-IV: when you only have a noisy proxy

- Proxy with measurement error: $z_t = \alpha + \beta\,\varepsilon_t + v_t \rightarrow$ LP of y on z attenuated
- **Unit-effect normalization:** attenuation cancels in the ratio

$$\theta_h = \frac{\theta_{z\to y,h}}{\theta_{z\to s,0}} \quad = \quad \text{2SLS of } y_{t+h} \text{ on } s_t, \text{ instrumented by } z_t$$

- **Controls:** if $z_t = \alpha + \beta\varepsilon_t + \psi' w_{t-1} + \cdots + v_t$, control for the lags $w_{t-1}, \ldots$; then HC SEs

LP-IV: when you only have a noisy proxy

- Proxy with measurement error: $z_t = \alpha + \beta\,\varepsilon_t + v_t \rightarrow$ LP of y on z attenuated
- **Unit-effect normalization:** attenuation cancels in the ratio

$$\theta_h = \frac{\theta_{z\to y,h}}{\theta_{z\to s,0}} \quad = \quad \text{2SLS of } y_{t+h} \text{ on } s_t, \text{ instrumented by } z_t$$

- **Controls:** if $z_t = \alpha + \beta\varepsilon_t + \psi' w_{t-1} + \cdots + v_t$, control for the lags $w_{t-1}, \ldots$; then HC SEs
- **What you need:**
  - Relevance $\beta \neq 0$: report first-stage F (running example: 52)
  - Lead-lag exogeneity: v_t uncorrelated with all shocks at all leads and lags
  - A strong normalization variable s_t (our 1-year rate)

LP-IV: when you only have a noisy proxy

- Proxy with measurement error: $z_t = \alpha + \beta\,\varepsilon_t + v_t \rightarrow$ LP of y on z attenuated
- **Unit-effect normalization:** attenuation cancels in the ratio

$$\theta_h = \frac{\theta_{z\to y,h}}{\theta_{z\to s,0}} \quad = \quad \text{2SLS of } y_{t+h} \text{ on } s_t, \text{ instrumented by } z_t$$

- **Controls:** if $z_t = \alpha + \beta\varepsilon_t + \psi' w_{t-1} + \cdots + v_t$, control for the lags $w_{t-1}, \ldots$; then HC SEs
- **What you need:**
  - Relevance $\beta \neq 0$: report first-stage F (running example: 52)
  - Lead-lag exogeneity: v_t uncorrelated with all shocks at all leads and lags
  - A strong normalization variable s_t (our 1-year rate)
- Narrative proxies work too: our z_t reads policy decisions off the Bank's records

[Plagborg-Møller (2026), lecture 1]

Why the ratio works, and when it fails

Why the ratio works, and when it fails

- Noise in z attenuates both LPs by the same factor $b \in (0, 1)$:

$$\theta_{z \to y,h} = b \times \theta_{\varepsilon \to y,h}, \qquad \theta_{z \to s,0} = b \times \theta_{\varepsilon \to s,0} \quad \Rightarrow \quad \frac{\theta_{z \to y,h}}{\theta_{z \to s,0}} = \frac{\theta_{\varepsilon \to y,h}}{\theta_{\varepsilon \to s,0}}$$

Why the ratio works, and when it fails

- Noise in z attenuates both LPs by the same factor $b \in (0, 1)$:

$$\theta_{z \to y,h} = b \times \theta_{\varepsilon \to y,h}, \qquad \theta_{z \to s,0} = b \times \theta_{\varepsilon \to s,0} \quad \Rightarrow \quad \frac{\theta_{z \to y,h}}{\theta_{z \to s,0}} = \frac{\theta_{\varepsilon \to y,h}}{\theta_{\varepsilon \to s,0}}$$

- **Proxy assumptions** for $\tilde{z}_t = \alpha + \beta \varepsilon_t + v_t$ (net of lagged controls):
  1. $\beta \neq 0$ (relevance)
  2. v_t uncorrelated with ε_t, y_t, s_t and other macro variables at all leads and lags

Why the ratio works, and when it fails

- Noise in z attenuates both LPs by the same factor $b \in (0, 1)$:

$$\theta_{z \to y,h} = b \times \theta_{\varepsilon \to y,h}, \qquad \theta_{z \to s,0} = b \times \theta_{\varepsilon \to s,0} \quad \Rightarrow \quad \frac{\theta_{z \to y,h}}{\theta_{z \to s,0}} = \frac{\theta_{\varepsilon \to y,h}}{\theta_{\varepsilon \to s,0}}$$

- **Proxy assumptions** for $\tilde{z}_t = \alpha + \beta \varepsilon_t + v_t$ (net of lagged controls):
  1. $\beta \neq 0$ (relevance)
  2. v_t uncorrelated with ε_t, y_t, s_t and other macro variables at all leads and lags

- **Weak instruments:** if z barely moves s on impact, LP-IV nearly divides by zero
  $\to$ median bias, under-covering bands. Two causes: z mostly noise, or s reacts weakly on impact

Why the ratio works, and when it fails

- Noise in z attenuates both LPs by the same factor $b \in (0, 1)$:

$$\theta_{z \to y,h} = b \times \theta_{\varepsilon \to y,h}, \qquad \theta_{z \to s,0} = b \times \theta_{\varepsilon \to s,0} \quad \Rightarrow \quad \frac{\theta_{z \to y,h}}{\theta_{z \to s,0}} = \frac{\theta_{\varepsilon \to y,h}}{\theta_{\varepsilon \to s,0}}$$

- **Proxy assumptions** for $\tilde{z}_t = \alpha + \beta \varepsilon_t + v_t$ (net of lagged controls):
  1. $\beta \neq 0$ (relevance)
  2. v_t uncorrelated with ε_t, y_t, s_t and other macro variables at all leads and lags

- **Weak instruments:** if z barely moves s on impact, LP-IV nearly divides by zero
  $\to$ median bias, under-covering bands. Two causes: z mostly noise, or s reacts weakly on impact

$\Rightarrow$ Pick a normalization variable that moves strongly with z. Proxy only: the FEVD is not identified; using z as if it were the shock understates it

[Plagborg-Møller (2026), lecture 1]

Where instruments come from

Where instruments come from

- **High-frequency surprises:** futures-implied rates in a 30-minute window around announcements
  - US [Kuttner (2001), Gürkaynak, Sack & Swanson (2005), Gertler & Karadi (2015)]
  - UK [Cesa-Bianchi, Thwaites & Vicondoa (2020)]
  - Euro area [Altavilla et al. (2019)]

Where instruments come from

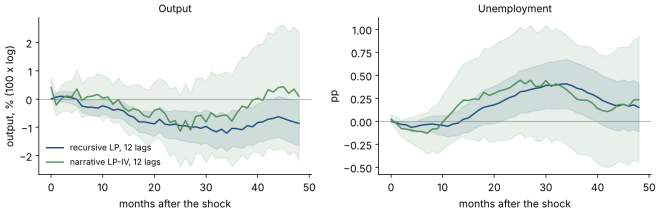
- **High-frequency surprises:** futures-implied rates in a 30-minute window around announcements
  - US [Kuttner (2001), Gürkaynak, Sack & Swanson (2005), Gertler & Karadi (2015)]
  - UK [Cesa-Bianchi, Thwaites & Vicondoa (2020)]
  - Euro area [Altavilla et al. (2019)]
- **Narrative series:** read the records to isolate changes not driven by the outlook
  - [Romer & Romer (2004, 2010), Ramey (2011), Känzig (2021)]
  - UK [Cloyne & Hürtgen (2016)] → our running example

Where instruments come from

- **High-frequency surprises:** futures-implied rates in a 30-minute window around announcements
  - US [Kuttner (2001), Gürkaynak, Sack & Swanson (2005), Gertler & Karadi (2015)]
  - UK [Cesa-Bianchi, Thwaites & Vicondoa (2020)]
  - Euro area [Altavilla et al. (2019)]
- **Narrative series:** read the records to isolate changes not driven by the outlook
  - [Romer & Romer (2004, 2010), Ramey (2011), Känzig (2021)]
  - UK [Cloyne & Hürtgen (2016)] → our running example
- **Known threats:**
  - Information effects: surprise reveals central bank's view [Nakamura & Steinsson (2018), Jarociński & Karadi (2020)]
  - Predictability: surprises correlate with past data [Bauer & Swanson (2023), Miranda-Agrippino & Ricco (2021)]

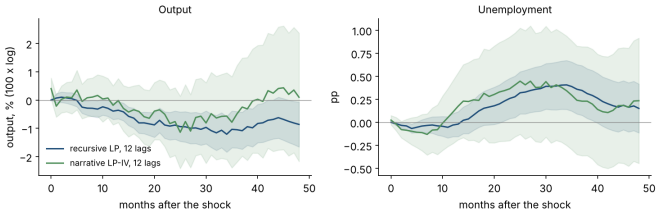
L3: the narrative shock as an instrument

1st: $i_t = \alpha + \beta z_t + \sum_{\ell=1}^{12} \psi_\ell' w_{t-\ell} + v_t$ 2nd: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_{\ell=1}^{12} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t+h}, \quad w_t = (y_t, u_t, i_t, z_t)'$



L3: the narrative shock as an instrument

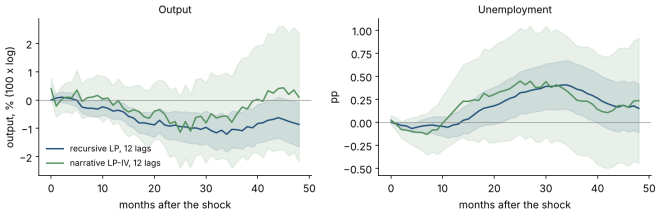
1st: $i_t = \alpha + \beta z_t + \sum_{\ell=1}^{12} \psi_\ell' w_{t-\ell} + v_t$ 2nd: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_{\ell=1}^{12} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t+h}, \quad w_t = (y_t, u_t, i_t, z_t)'$



- First-stage $F = 52$; no date-t controls: the instrument does the identifying

L3: the narrative shock as an instrument

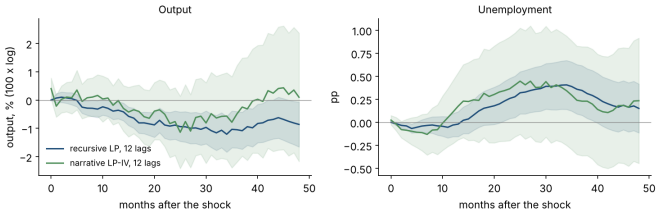
1st: $i_t = \alpha + \beta z_t + \sum_{\ell=1}^{12} \psi_\ell' w_{t-\ell} + v_t$ 2nd: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_{\ell=1}^{12} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t+h}, \quad w_t = (y_t, u_t, i_t, z_t)'$



- First-stage $F = 52$; no date-t controls: the instrument does the identifying
- Output: -0.18% at 12 months, -0.76% at 24 (se 0.42, 0.73); unemployment $+0.41$pp at 24

L3: the narrative shock as an instrument

1st: $i_t = \alpha + \beta z_t + \sum_{\ell=1}^{12} \psi_\ell' w_{t-\ell} + v_t$ 2nd: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_{\ell=1}^{12} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t+h}, \quad w_t = (y_t, u_t, i_t, z_t)'$

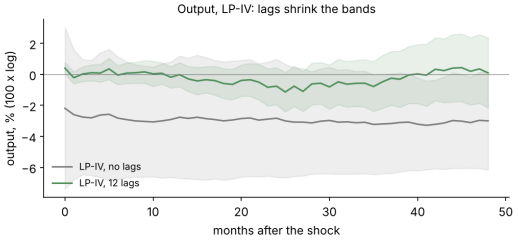


- First-stage $F = 52$; no date-t controls: the instrument does the identifying
- Output: -0.18% at 12 months, -0.76% at 24 (se 0.42, 0.73); unemployment $+0.41$pp at 24

$\Rightarrow$ A different assumption gives a similar answer at 2 years, with wider, more honest bands

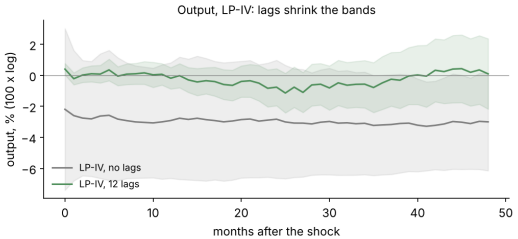
L4: lags still matter with an instrument

$$y_{t+h} = \mu_h + \theta_h \hat{i}_t + \xi_{h,t+h}, \qquad \hat{i}_t \text{ from } i_t = \alpha + \beta z_t + v_t \qquad \text{(L3 without the lags)}$$



L4: lags still matter with an instrument

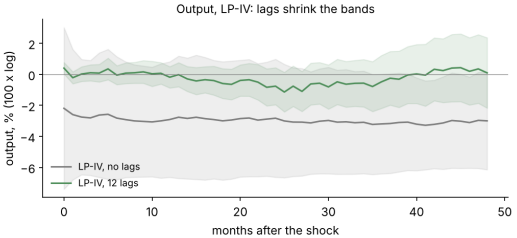
$$y_{t+h} = \mu_h + \theta_h \hat{i}_t + \xi_{h,t+h}, \qquad \hat{i}_t \text{ from } i_t = \alpha + \beta z_t + v_t \qquad \text{(L3 without the lags)}$$



- No lags: -2.9% at every horizon, se 2.3 $\rightarrow$ the trend again, and 5 times the standard error

L4: lags still matter with an instrument

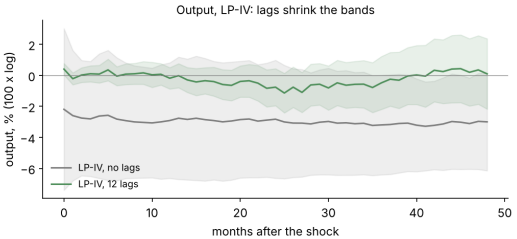
$$y_{t+h} = \mu_h + \theta_h \hat{i}_t + \xi_{h,t+h}, \qquad \hat{i}_t \text{ from } i_t = \alpha + \beta z_t + v_t \qquad \text{(L3 without the lags)}$$



- No lags: -2.9% at every horizon, se 2.3 $\rightarrow$ the trend again, and 5 times the standard error
- z_t is partly predictable: $R^2 = 0.29$ on 12 lags of the data

L4: lags still matter with an instrument

$$y_{t+h} = \mu_h + \theta_h \hat{i}_t + \xi_{h,t+h}, \qquad \hat{i}_t \text{ from } i_t = \alpha + \beta z_t + v_t \qquad \text{(L3 without the lags)}$$

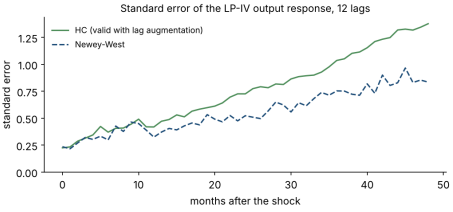


- No lags: -2.9% at every horizon, se 2.3 $\rightarrow$ the trend again, and 5 times the standard error
- z_t is partly predictable: $R^2 = 0.29$ on 12 lags of the data

$\Rightarrow$ Lags soak up residual variance and the predictable part of z_t: always include them

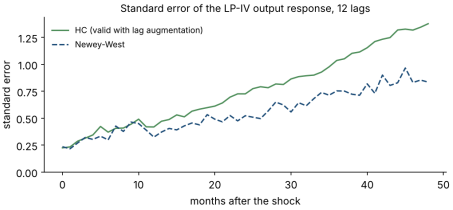
L5: which standard errors?

$$\hat{V} = (\hat{X}'\hat{X})^{-1}\hat{\Omega}\,(\hat{X}'\hat{X})^{-1}, \quad s_t = \hat{X}_t\hat{\xi}_{h,t+h}: \quad \hat{\Omega}_{HC} = \sum_t s_t s_t', \quad \hat{\Omega}_{NW} = \hat{\Omega}_{HC} + \sum_{\ell=1}^{L}\left(1 - \tfrac{\ell}{L+1}\right)\sum_t \left(s_t s_{t-\ell}' + s_{t-\ell}s_t'\right)$$



L5: which standard errors?

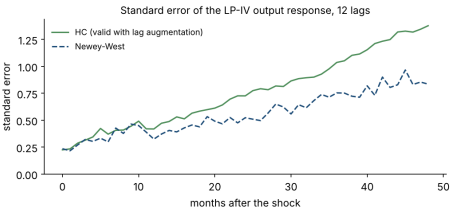
$$\hat{V} = (\hat{X}'\hat{X})^{-1}\hat{\Omega}\,(\hat{X}'\hat{X})^{-1}, \quad s_t = \hat{X}_t\hat{\xi}_{h,t+h}: \quad \hat{\Omega}_{HC} = \textstyle\sum_t s_t s_t', \quad \hat{\Omega}_{NW} = \hat{\Omega}_{HC} + \sum_{\ell=1}^{L}\left(1 - \frac{\ell}{L+1}\right)\sum_t \left(s_t s_{t-\ell}' + s_{t-\ell}s_t'\right)$$



- LP-IV of L3, $L = h + 5$. At 12, 24, 36 months: HC 0.42, 0.73, 1.04; Newey-West 0.32, 0.52, 0.75

L5: which standard errors?

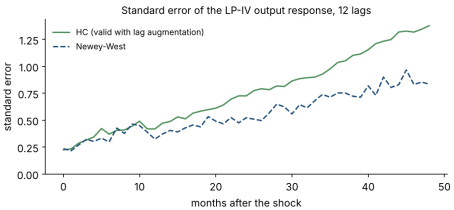
$$\hat{V} = (\hat{X}'\hat{X})^{-1}\hat{\Omega}\,(\hat{X}'\hat{X})^{-1}, \quad s_t = \hat{X}_t\hat{\xi}_{h,t+h}: \quad \hat{\Omega}_{HC} = \textstyle\sum_t s_t s_t', \quad \hat{\Omega}_{NW} = \hat{\Omega}_{HC} + \sum_{\ell=1}^{L}\left(1 - \frac{\ell}{L+1}\right)\sum_t \left(s_t s_{t-\ell}' + s_{t-\ell}s_t'\right)$$



- LP-IV of L3, $L = h + 5$. At 12, 24, 36 months: HC 0.42, 0.73, 1.04; Newey-West 0.32, 0.52, 0.75
- Newey-West is about 30% narrower beyond a year, where it is known to under-cover

L5: which standard errors?

$$\hat{V} = (\hat{X}'\hat{X})^{-1}\hat{\Omega}\,(\hat{X}'\hat{X})^{-1}, \quad s_t = \hat{X}_t\hat{\xi}_{h,t+h}: \quad \hat{\Omega}_{HC} = \textstyle\sum_t s_t s_t', \quad \hat{\Omega}_{NW} = \hat{\Omega}_{HC} + \sum_{\ell=1}^{L}\left(1 - \frac{\ell}{L+1}\right)\sum_t \left(s_t s_{t-\ell}' + s_{t-\ell}s_t'\right)$$



- LP-IV of L3, $L = h + 5$. At 12, 24, 36 months: HC 0.42, 0.73, 1.04; Newey-West 0.32, 0.52, 0.75
- Newey-West is about 30% narrower beyond a year, where it is known to under-cover

$\Rightarrow$ With lags as controls, plain HC standard errors are valid: use them

Jagged responses: smoothing and shrinkage

Jagged responses: smoothing and shrinkage

- LP has no structure across horizons $\rightarrow$ jagged $\hat{\theta}_h$: the variance side of low bias

Jagged responses: smoothing and shrinkage

- LP has no structure across horizons $\rightarrow$ jagged $\hat{\theta}_h$: the variance side of low bias
- **Parametric fit:** regress $\hat{\theta}_h$ on a polynomial in h, e.g. $f(h) = a_0 + a_1 h + a_2 h^2$

  Jordà & Taylor (2025): fit the log of the responses if they share a sign

Jagged responses: smoothing and shrinkage

- LP has no structure across horizons $\rightarrow$ jagged $\hat{\theta}_h$: the variance side of low bias
- **Parametric fit:** regress $\hat{\theta}_h$ on a polynomial in h, e.g. $f(h) = a_0 + a_1 h + a_2 h^2$

  Jordà & Taylor (2025): fit the log of the responses if they share a sign
- **Shrink part of the way**, more when estimates are noisy or the fit is good:

$$\hat{\theta}_h^{smooth} = (1 - c)\,\hat{\theta}_h + c\,\hat{\theta}_h^{fit}, \qquad c = \min\left\{\frac{\text{tr(noise)}}{\text{SSR of fit}}, 1\right\}$$

[Plagborg-Møller (2016); penalized LP: Barnichon & Brownlees (2019)]

Jagged responses: smoothing and shrinkage

- LP has no structure across horizons $\rightarrow$ jagged $\hat{\theta}_h$: the variance side of low bias
- **Parametric fit:** regress $\hat{\theta}_h$ on a polynomial in h, e.g. $f(h) = a_0 + a_1 h + a_2 h^2$

  Jordà & Taylor (2025): fit the log of the responses if they share a sign
- **Shrink part of the way**, more when estimates are noisy or the fit is good:

$$\hat{\theta}_h^{smooth} = (1 - c)\,\hat{\theta}_h + c\,\hat{\theta}_h^{fit}, \qquad c = \min\left\{\frac{\text{tr(noise)}}{\text{SSR of fit}}, 1\right\}$$

  [Plagborg-Møller (2016); penalized LP: Barnichon & Brownlees (2019)]

$\Rightarrow$ A visual aid: always report the band of the original LP, not one around the smoothed line

[Plagborg-Møller (2026), lecture 3]

L6: LP variance decomposition

L6: LP variance decomposition

- R^2 **method** [Gorodnichenko & Lee (2020)]
  1. $\tilde{x}_t$: residual of x_t on all controls
  2. $\hat{f}_{h,t+h}$: residual of y_{t+h} on the same controls
  3. $FEVD_h = R^2$ of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$

L6: LP variance decomposition

- R^2 **method** [Gorodnichenko & Lee (2020)]
  1. $\tilde{x}_t$: residual of x_t on all controls
  2. $\hat{f}_{h,t+h}$: residual of y_{t+h} on the same controls
  3. $FEVD_h = R^2$ of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$

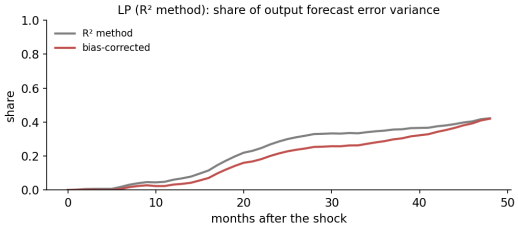
- **Small-sample bias** ($h+1$ regressors): simulate from a fitted VAR with known FEVD; subtract mean LP R^2 − true FEVD

L6: LP variance decomposition

- R^2 **method** [Gorodnichenko & Lee (2020)]
  1. $\tilde{x}_t$: residual of x_t on all controls
  2. $\hat{f}_{h,t+h}$: residual of y_{t+h} on the same controls
  3. $FEVD_h = R^2$ of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$
- **Small-sample bias** ($h+1$ regressors): simulate from a fitted VAR with known FEVD; subtract mean LP R^2 − true FEVD
- Noisy proxy only → FEVD not point-identified [bounds: Plagborg-Møller & Wolf (2022)]

L6: LP variance decomposition

- **R^2 method** [Gorodnichenko & Lee (2020)]
  1. $\tilde{x}_t$: residual of x_t on all controls
  2. $\hat{f}_{h,t+h}$: residual of y_{t+h} on the same controls
  3. $FEVD_h = R^2$ of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$
- **Small-sample bias** ($h+1$ regressors): simulate from a fitted VAR with known FEVD; subtract mean LP R^2 − true FEVD
- Noisy proxy only → FEVD not point-identified [bounds: Plagborg-Møller & Wolf (2022)]



Running example, $\tilde{x}_t$ from L2, at 12 / 24 / 48 months: raw 5% / 27% / 42%, bias-corrected 2% / 20% / 41%

Why applied work loves LPs: easy extensions

Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]

Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]
- **Panels:** fixed effects, one regression per h [Jordà, Schularick & Taylor (2013)]

Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]
- **Panels:** fixed effects, one regression per h [Jordà, Schularick & Taylor (2013)]
- **Multipliers:** regress $\sum \Delta Y$ on $\sum \Delta G$, instrumented by the shock [Ramey & Zubairy (2018)]

Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]
- **Panels:** fixed effects, one regression per h [Jordà, Schularick & Taylor (2013)]
- **Multipliers:** regress $\sum \Delta Y$ on $\sum \Delta G$, instrumented by the shock [Ramey & Zubairy (2018)]
- **Long differences:** outcome $y_{t+h} - y_{t-1}$ for trending series [Piger & Stockwell (2025)]

Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]
- **Panels:** fixed effects, one regression per h [Jordà, Schularick & Taylor (2013)]
- **Multipliers:** regress $\sum \Delta Y$ on $\sum \Delta G$, instrumented by the shock [Ramey & Zubairy (2018)]
- **Long differences:** outcome $y_{t+h} - y_{t-1}$ for trending series [Piger & Stockwell (2025)]
- **Asymmetry and size:** separate coefficients on positive/negative, large/small shocks

Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]
- **Panels:** fixed effects, one regression per h [Jordà, Schularick & Taylor (2013)]
- **Multipliers:** regress $\sum \Delta Y$ on $\sum \Delta G$, instrumented by the shock [Ramey & Zubairy (2018)]
- **Long differences:** outcome $y_{t+h} - y_{t-1}$ for trending series [Piger & Stockwell (2025)]
- **Asymmetry and size:** separate coefficients on positive/negative, large/small shocks
- Caution:
  - Nonlinear LPs need care if the state responds to the shock [Gonçalves et al. (2024)]
  - LP also has small-sample bias with persistent data [Herbst & Johannsen (2024)]

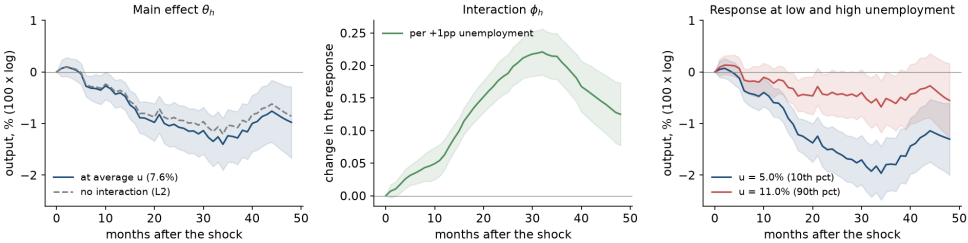
Why applied work loves LPs: easy extensions

- **State dependence:** interact the shock with a lagged state [Auerbach & Gorodnichenko (2013), Ramey & Zubairy (2018)]
- **Panels:** fixed effects, one regression per h [Jordà, Schularick & Taylor (2013)]
- **Multipliers:** regress $\sum \Delta Y$ on $\sum \Delta G$, instrumented by the shock [Ramey & Zubairy (2018)]
- **Long differences:** outcome $y_{t+h} - y_{t-1}$ for trending series [Piger & Stockwell (2025)]
- **Asymmetry and size:** separate coefficients on positive/negative, large/small shocks
- Caution:
  - Nonlinear LPs need care if the state responds to the shock [Gonçalves et al. (2024)]
  - LP also has small-sample bias with persistent data [Herbst & Johannsen (2024)]

$\Rightarrow$ Coming back in Weeks 3–5: crises, open-economy panels, regional LPs

L7: state dependence, interact the shock with the state

$$y_{t+h} = \mu_h + \theta_h\, i_t + \phi_h\, i_t\,(u_{t-1} - \bar{u}) + \gamma_h'(y_t, u_t) + \sum_{\ell=1}^{12} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$$



L7: state dependence, interact the shock with the state

$$y_{t+h} = \mu_h + \theta_h\, i_t + \phi_h\, i_t\,(u_{t-1} - \bar{u}) + \gamma_h'(y_t, u_t) + \sum_{\ell=1}^{12} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$$



- Response at $u = c$: $\theta_h + \phi_h(c - \bar{u}) \rightarrow$ center the state at c, read off the coefficient on i_t and its SE

L7: state dependence, interact the shock with the state

$$y_{t+h} = \mu_h + \theta_h\, i_t + \phi_h\, i_t\,(u_{t-1} - \bar{u}) + \gamma_h'(y_t, u_t) + \textstyle\sum_{\ell=1}^{12} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$$



- Response at $u = c$: $\theta_h + \phi_h(c - \bar{u}) \to$ center the state at c, read off the coefficient on i_t and its SE
- 24 months: -1.49% at $u = 5\%$, -0.40% at $u = 11\%$; narrative IV too weak here (first-stage $t = 1.8$)

L7: state dependence, interact the shock with the state

$$y_{t+h} = \mu_h + \theta_h\, i_t + \phi_h\, i_t\,(u_{t-1} - \bar{u}) + \gamma_h'(y_t, u_t) + \sum_{\ell=1}^{12} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$$



- Response at $u = c$: $\theta_h + \phi_h(c - \bar{u}) \to$ center the state at c, read off the coefficient on i_t and its SE
- 24 months: -1.49% at $u = 5\%$, -0.40% at $u = 11\%$; narrative IV too weak here (first-stage $t = 1.8$)

$\Rightarrow$ Lagged state; high $u \approx$ the 1980s; careful if the shock moves the state [Gonçalves et al. (2024)]

Cumulative responses and multipliers

Cumulative responses and multipliers

- **Cumulative IRF:** put $\sum_{j=0}^{h} y_{t+j}$ on the left $\rightarrow$ one LP gives $\sum_{j \leq h} \theta_j$ with its SE

Cumulative responses and multipliers

- **Cumulative IRF:** put $\sum_{j=0}^{h} y_{t+j}$ on the left $\rightarrow$ one LP gives $\sum_{j \leq h} \theta_j$ with its SE
- **Multiplier:** 2SLS per horizon, instrument: the shock

$$\sum_{j=0}^{h} y_{t+j} = \gamma_h + \phi_h(L)\, w_{t-1} + m_h \sum_{j=0}^{h} g_{t+j} + \omega_{t+h}$$

Cumulative responses and multipliers

- **Cumulative IRF:** put $\sum_{j=0}^{h} y_{t+j}$ on the left $\rightarrow$ one LP gives $\sum_{j \leq h} \theta_j$ with its SE
- **Multiplier:** 2SLS per horizon, instrument: the shock

$$\sum_{j=0}^{h} y_{t+j} = \gamma_h + \phi_h(L)\, w_{t-1} + m_h \sum_{j=0}^{h} g_{t+j} + \omega_{t+h}$$

- $= \sum_j \theta_j^y / \sum_j \theta_j^g$ on the same sample, but with the SE of m_h, a first-stage F by horizon, and several shocks at once

Cumulative responses and multipliers

- **Cumulative IRF:** put $\sum_{j=0}^{h} y_{t+j}$ on the left $\rightarrow$ one LP gives $\sum_{j\leq h} \theta_j$ with its SE
- **Multiplier:** 2SLS per horizon, instrument: the shock

$$\sum_{j=0}^{h} y_{t+j} = \gamma_h + \phi_h(L)\, w_{t-1} + m_h \sum_{j=0}^{h} g_{t+j} + \omega_{t+h}$$

- $= \sum_j \theta_j^y / \sum_j \theta_j^g$ on the same sample, but with the SE of m_h, a first-stage F by horizon, and several shocks at once
- **Pitfalls avoided:** peak over impact [Blanchard & Perotti (2002)] overstates; logs $\times$ mean Y/G are biased (Y/G: 2–24 since 1889) $\rightarrow$ divide Y, G by trend GDP

Cumulative responses and multipliers

- **Cumulative IRF:** put $\sum_{j=0}^{h} y_{t+j}$ on the left $\rightarrow$ one LP gives $\sum_{j\leq h} \theta_j$ with its SE
- **Multiplier:** 2SLS per horizon, instrument: the shock

$$\sum_{j=0}^{h} y_{t+j} = \gamma_h + \phi_h(L)\, w_{t-1} + m_h \sum_{j=0}^{h} g_{t+j} + \omega_{t+h}$$

- $= \sum_j \theta_j^y / \sum_j \theta_j^g$ on the same sample, but with the SE of m_h, a first-stage F by horizon, and several shocks at once
- **Pitfalls avoided:** peak over impact [Blanchard & Perotti (2002)] overstates; logs $\times$ mean Y/G are biased (Y/G: 2–24 since 1889) $\rightarrow$ divide Y, G by trend GDP
- **US 1889–2015:** $m_h \approx 0.6$–0.7, no larger in slack; above 1 only at the ZLB excluding WWII

Cumulative responses and multipliers

- **Cumulative IRF:** put $\sum_{j=0}^{h} y_{t+j}$ on the left $\rightarrow$ one LP gives $\sum_{j\leq h} \theta_j$ with its SE
- **Multiplier:** 2SLS per horizon, instrument: the shock

$$\sum_{j=0}^{h} y_{t+j} = \gamma_h + \phi_h(L)\, w_{t-1} + m_h \sum_{j=0}^{h} g_{t+j} + \omega_{t+h}$$

- $= \sum_j \theta_j^y / \sum_j \theta_j^g$ on the same sample, but with the SE of m_h, a first-stage F by horizon, and several shocks at once
- **Pitfalls avoided:** peak over impact [Blanchard & Perotti (2002)] overstates; logs $\times$ mean Y/G are biased (Y/G: 2–24 since 1889) $\rightarrow$ divide Y, G by trend GDP
- **US 1889–2015:** $m_h \approx 0.6$–0.7, no larger in slack; above 1 only at the ZLB excluding WWII

$\Rightarrow$ Monetary analogue $\sum y$ on $\sum i$ needs a persistent rate path: if it reverses, $\sum i \approx 0$ and the IV turns weak

[Ramey & Zubairy (2018)]

Structural VARs

Origins: Sims (1980)

Origins: Sims (1980)

- **The problem:** large Keynesian models identified by "incredible" exclusion restrictions

Origins: Sims (1980)

- **The problem:** large Keynesian models identified by "incredible" exclusion restrictions
- **The proposal:**
  - Treat every variable as endogenous, estimate a small unrestricted VAR
  - Impose assumptions only where needed: on the shocks

Origins: Sims (1980)

- **The problem:** large Keynesian models identified by "incredible" exclusion restrictions
- **The proposal:**
  - Treat every variable as endogenous, estimate a small unrestricted VAR
  - Impose assumptions only where needed: on the shocks
- **The legacy:** standard tool for monetary policy, fiscal policy and DSGE validation

  [Christiano, Eichenbaum & Evans (1999), Blanchard & Perotti (2002)]

Origins: Sims (1980)

- **The problem:** large Keynesian models identified by "incredible" exclusion restrictions
- **The proposal:**
  - Treat every variable as endogenous, estimate a small unrestricted VAR
  - Impose assumptions only where needed: on the shocks
- **The legacy:** standard tool for monetary policy, fiscal policy and DSGE validation
  [Christiano, Eichenbaum & Evans (1999), Blanchard & Perotti (2002)]

$\Rightarrow$ Identification moved from the equations to the shocks – the same frame as our LPs

The reduced-form VAR

The reduced-form VAR

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

The reduced-form VAR

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

- y_t is $k \times 1$; u_t are one-step-ahead forecast errors – correlated, not structural

The reduced-form VAR

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

- y_t is $k \times 1$; u_t are one-step-ahead forecast errors – correlated, not structural
- **Estimate:** equation-by-equation OLS (same regressors $\Rightarrow$ OLS = GLS)

The reduced-form VAR

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

- y_t is $k \times 1$; u_t are one-step-ahead forecast errors – correlated, not structural
- **Estimate:** equation-by-equation OLS (same regressors $\Rightarrow$ OLS = GLS)
- **Stack:** companion form $Y_t = FY_{t-1} + U_t$; stable if eigenvalues of F inside unit circle

The reduced-form VAR

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

- y_t is $k \times 1$; u_t are one-step-ahead forecast errors – correlated, not structural
- **Estimate:** equation-by-equation OLS (same regressors $\Rightarrow$ OLS = GLS)
- **Stack:** companion form $Y_t = FY_{t-1} + U_t$; stable if eigenvalues of F inside unit circle
- **Invert:** Wold MA representation, computed by iterating the VAR

$$y_t = \mu + \sum_{h=0}^{\infty} C_h u_{t-h}, \qquad C_0 = I, \quad C_h = \sum_{\ell=1}^{\min(h,p)} A_\ell \, C_{h-\ell} \quad \left(= [F^h]_{1:k,1:k} \right)$$

The reduced-form VAR

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

- y_t is $k \times 1$; u_t are one-step-ahead forecast errors – correlated, not structural
- **Estimate:** equation-by-equation OLS (same regressors $\Rightarrow$ OLS = GLS)
- **Stack:** companion form $Y_t = FY_{t-1} + U_t$; stable if eigenvalues of F inside unit circle
- **Invert:** Wold MA representation, computed by iterating the VAR

$$y_t = \mu + \sum_{h=0}^{\infty} C_h u_{t-h}, \qquad C_0 = I, \quad C_h = \sum_{\ell=1}^{\min(h,p)} A_\ell \, C_{h-\ell} \quad \left(= [F^h]_{1:k,1:k} \right)$$

$\Rightarrow$ C_h = reduced-form responses to forecast errors: no economic meaning yet

Choosing the lag length: the trade-off

Choosing the lag length: the trade-off

- **Question:** how many months of the past does the VAR need?
  - Too few lags: dynamics left in the residuals $\rightarrow$ biased responses, worst at long horizons
  - Too many lags: pk^2 coefficients (108 for $p = 12$, $k = 3$, with $T = 356$) $\rightarrow$ noisy estimates

Choosing the lag length: the trade-off

- **Question:** how many months of the past does the VAR need?
  - Too few lags: dynamics left in the residuals $\rightarrow$ biased responses, worst at long horizons
  - Too many lags: pk^2 coefficients (108 for $p = 12$, $k = 3$, with $T = 356$) $\rightarrow$ noisy estimates
- In-sample fit always improves with more lags (like R^2 with more regressors) $\rightarrow$ fit alone cannot choose

Choosing the lag length: the trade-off

- **Question:** how many months of the past does the VAR need?
  - Too few lags: dynamics left in the residuals $\rightarrow$ biased responses, worst at long horizons
  - Too many lags: pk^2 coefficients (108 for $p = 12$, $k = 3$, with $T = 356$) $\rightarrow$ noisy estimates
- In-sample fit always improves with more lags (like R^2 with more regressors) $\rightarrow$ fit alone cannot choose
- **Information criterion (IC)** = a score for each p: fit + penalty for the number of parameters

$$\text{one variable, AR}(p): \qquad IC(p) = \underbrace{\ln \hat{\sigma}_p^2}_{\text{fit: residual variance}} + \underbrace{c_T \frac{p}{T}}_{\text{penalty}}$$

Choosing the lag length: the trade-off

- **Question:** how many months of the past does the VAR need?
  - Too few lags: dynamics left in the residuals $\rightarrow$ biased responses, worst at long horizons
  - Too many lags: pk^2 coefficients (108 for $p = 12$, $k = 3$, with $T = 356$) $\rightarrow$ noisy estimates
- In-sample fit always improves with more lags (like R^2 with more regressors) $\rightarrow$ fit alone cannot choose
- **Information criterion (IC)** = a score for each p: fit + penalty for the number of parameters

$$\text{one variable, AR}(p): \qquad IC(p) = \underbrace{\ln \hat{\sigma}_p^2}_{\text{fit: residual variance}} + \underbrace{c_T \frac{p}{T}}_{\text{penalty}}$$

- **Rule:** estimate $p = 1, \ldots, p_{max}$ on the same sample, pick the p with the lowest IC

Choosing the lag length: the trade-off

- **Question:** how many months of the past does the VAR need?
  - Too few lags: dynamics left in the residuals $\rightarrow$ biased responses, worst at long horizons
  - Too many lags: pk^2 coefficients (108 for $p = 12$, $k = 3$, with $T = 356$) $\rightarrow$ noisy estimates
- In-sample fit always improves with more lags (like R^2 with more regressors) $\rightarrow$ fit alone cannot choose
- **Information criterion (IC)** = a score for each p: fit + penalty for the number of parameters

$$\text{one variable, AR}(p): \qquad IC(p) = \underbrace{\ln \hat{\sigma}_p^2}_{\text{fit: residual variance}} + \underbrace{c_T \frac{p}{T}}_{\text{penalty}}$$

- **Rule:** estimate $p = 1, \ldots, p_{max}$ on the same sample, pick the p with the lowest IC
- Standard tool: Python `statsmodels` `VAR(...).select_order()` [Akaike (1974), Schwarz (1978)]

Choosing the lag length: information criteria for a VAR

Choosing the lag length: information criteria for a VAR

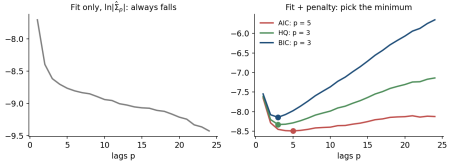
$$IC(p) = \underbrace{\ln|\hat{\Sigma}_p|}_{\text{fit}} + \underbrace{c_T \frac{pk^2}{T}}_{\text{penalty}}, \qquad c_T = 2 \text{ (AIC)},\ 2\ln\ln T \approx 3.5 \text{ (HQ)},\ \ln T \approx 5.8 \text{ (BIC)}$$

- $|\hat{\Sigma}_p|$ = size of all residuals jointly ($k = 1$: $\hat{\sigma}^2$). Compare across p within one criterion; fit alone always falls

Choosing the lag length: information criteria for a VAR

$$IC(p) = \underbrace{\ln|\hat{\Sigma}_p|}_{\text{fit}} + \underbrace{c_T \frac{pk^2}{T}}_{\text{penalty}}, \qquad c_T = 2 \text{ (AIC)},\ 2\ln\ln T \approx 3.5 \text{ (HQ)},\ \ln T \approx 5.8 \text{ (BIC)}$$

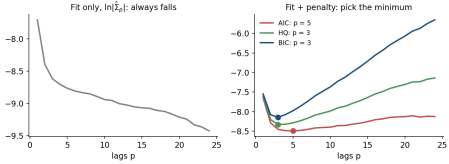
- $|\hat{\Sigma}_p|$ = size of all residuals jointly ($k = 1$: $\hat{\sigma}^2$). Compare across p within one criterion; fit alone always falls



Choosing the lag length: information criteria for a VAR

$$IC(p) = \underbrace{\ln|\hat{\Sigma}_p|}_{\text{fit}} + \underbrace{c_T \frac{pk^2}{T}}_{\text{penalty}}, \qquad c_T = 2 \text{ (AIC)},\ 2\ln\ln T \approx 3.5 \text{ (HQ)},\ \ln T \approx 5.8 \text{ (BIC)}$$

- $|\hat{\Sigma}_p|$ = size of all residuals jointly ($k = 1$: $\hat{\sigma}^2$). Compare across p within one criterion; fit alone always falls

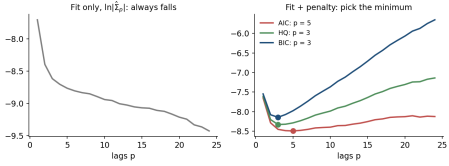


- **UK:** AIC $p = 5$, HQ and BIC $p = 3$, yet we use $p = 12$. ICs target 1-month forecasts, we want 48-month responses; missing lags cause bias that grows with h (C1)

Choosing the lag length: information criteria for a VAR

$$IC(p) = \underbrace{\ln|\hat{\Sigma}_p|}_{\text{fit}} + \underbrace{c_T \frac{pk^2}{T}}_{\text{penalty}}, \qquad c_T = 2 \text{ (AIC)},\ 2\ln\ln T \approx 3.5 \text{ (HQ)},\ \ln T \approx 5.8 \text{ (BIC)}$$

- $|\hat{\Sigma}_p|$ = size of all residuals jointly ($k = 1$: $\hat{\sigma}^2$). Compare across p within one criterion; fit alone always falls



- **UK:** AIC $p = 5$, HQ and BIC $p = 3$, yet we use $p = 12$. ICs target 1-month forecasts, we want 48-month responses; missing lags cause bias that grows with h (C1)

$\Rightarrow$ ICs are a minimum; with monthly data use at least 12 lags

The identification problem

The identification problem

- **What the VAR gives us:** forecast errors u_t = this month's surprises in output, unemployment and the rate

The identification problem

- **What the VAR gives us:** forecast errors u_t = this month's surprises in output, unemployment and the rate
- **What we want:** the policy shock: a rate move that is not a response to anything else

The identification problem

- **What the VAR gives us:** forecast errors u_t = this month's surprises in output, unemployment and the rate
- **What we want:** the policy shock: a rate move that is not a response to anything else
- **The problem:** a surprise rate rise can be
  - a policy shock, or
  - the Bank reacting to news about output it saw this month

  The surprises are correlated, so the rate surprise mixes the two

The identification problem

- **What the VAR gives us:** forecast errors u_t = this month's surprises in output, unemployment and the rate
- **What we want:** the policy shock: a rate move that is not a response to anything else
- **The problem:** a surprise rate rise can be
  - a policy shock, or
  - the Bank reacting to news about output it saw this month

  The surprises are correlated, so the rate surprise mixes the two

$\Rightarrow$ Identification = strip the reaction out of the rate surprise. Simplest way: a regression

The simple idea: regress one surprise on the other

The simple idea: regress one surprise on the other

Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

$$u_{i,t} = \gamma\, u_{y,t} + e_t$$

The simple idea: regress one surprise on the other

Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

$$u_{i,t} = \gamma\, u_{y,t} + e_t$$

- $\hat{\gamma} = \frac{\text{Cov}(u_y, u_i)}{\text{Var}(u_y)} = \frac{0.010}{0.246} = 0.04$: the Bank's reaction to an output surprise within the month

The simple idea: regress one surprise on the other

Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

$$u_{i,t} = \gamma\, u_{y,t} + e_t$$

- $\hat{\gamma} = \frac{\text{Cov}(u_y, u_i)}{\text{Var}(u_y)} = \frac{0.010}{0.246} = 0.04$: the Bank's reaction to an output surprise within the month
- Residual e_t = the part of the rate surprise output cannot explain = the policy shock
  s.d. $= \sqrt{0.127 - 0.010^2/0.246} = 0.36$pp

The simple idea: regress one surprise on the other

Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

$$u_{i,t} = \gamma\, u_{y,t} + e_t$$

- $\hat{\gamma} = \frac{\text{Cov}(u_y, u_i)}{\text{Var}(u_y)} = \frac{0.010}{0.246} = 0.04$: the Bank's reaction to an output surprise within the month
- Residual e_t = the part of the rate surprise output cannot explain = the policy shock
  s.d. $= \sqrt{0.127 - 0.010^2/0.246} = 0.36$pp
- **Assumption:** the Bank may react to output within the month, but output does not react to the rate within the month (else $\hat{\gamma}$ mixes both directions)

The simple idea: regress one surprise on the other

Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

$$u_{i,t} = \gamma\, u_{y,t} + e_t$$

- $\hat{\gamma} = \frac{\mathrm{Cov}(u_y, u_i)}{\mathrm{Var}(u_y)} = \frac{0.010}{0.246} = 0.04$: the Bank's reaction to an output surprise within the month
- Residual e_t = the part of the rate surprise output cannot explain = the policy shock
  s.d. $= \sqrt{0.127 - 0.010^2/0.246} = 0.36$pp
- **Assumption:** the Bank may react to output within the month, but output does not react to the rate within the month (else $\hat{\gamma}$ mixes both directions)
- Run it the other way (u_y on u_i) and you assume the opposite. Both fit the data equally well: the data cannot tell which way causality runs $\rightarrow$ the direction is the economics

The simple idea: regress one surprise on the other

Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

$$u_{i,t} = \gamma\, u_{y,t} + e_t$$

- $\hat{\gamma} = \frac{\text{Cov}(u_y, u_i)}{\text{Var}(u_y)} = \frac{0.010}{0.246} = 0.04$: the Bank's reaction to an output surprise within the month
- Residual e_t = the part of the rate surprise output cannot explain = the policy shock
  s.d. $= \sqrt{0.127 - 0.010^2/0.246} = 0.36$pp
- **Assumption:** the Bank may react to output within the month, but output does not react to the rate within the month (else $\hat{\gamma}$ mixes both directions)
- Run it the other way (u_y on u_i) and you assume the opposite. Both fit the data equally well: the data cannot tell which way causality runs → the direction is the economics

⇒ Same idea as the LP with date-t controls (L2): the rate, net of this month's output

The same regression in matrix form

The same regression in matrix form

Two surprises ($u_{y,t}, u_{i,t}$), two shocks (demand $\varepsilon_{d,t}$, policy $\varepsilon_{m,t}$), variance 1

The same regression in matrix form

Two surprises $(u_{y,t}, u_{i,t})$, two shocks (demand $\varepsilon_{d,t}$, policy $\varepsilon_{m,t}$), variance 1

Unknowns:

$$\begin{pmatrix} u_{y,t} \\ u_{i,t} \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\Theta_0} \begin{pmatrix} \varepsilon_{d,t} \\ \varepsilon_{m,t} \end{pmatrix}$$

- One entry per (variable, shock) $\rightarrow k \times k = 4$
- b: policy $\rightarrow$ output; c: demand $\rightarrow$ rate

The same regression in matrix form

Two surprises $(u_{y,t}, u_{i,t})$, two shocks (demand $\varepsilon_{d,t}$, policy $\varepsilon_{m,t}$), variance 1

Unknowns:

$$\begin{pmatrix} u_{y,t} \\ u_{i,t} \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\Theta_0} \begin{pmatrix} \varepsilon_{d,t} \\ \varepsilon_{m,t} \end{pmatrix}$$

Data: the covariance matrix of the surprises

$$\hat{\Sigma} = \begin{pmatrix} 0.246 & 0.010 \\ 0.010 & 0.127 \end{pmatrix}$$

- One entry per (variable, shock) $\rightarrow$ $k \times k = 4$
- b: policy $\rightarrow$ output; c: demand $\rightarrow$ rate

The same regression in matrix form

Two surprises ($u_{y,t}, u_{i,t}$), two shocks (demand $\varepsilon_{d,t}$, policy $\varepsilon_{m,t}$), variance 1

Unknowns:

$$\begin{pmatrix} u_{y,t} \\ u_{i,t} \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\Theta_0} \begin{pmatrix} \varepsilon_{d,t} \\ \varepsilon_{m,t} \end{pmatrix}$$

- One entry per (variable, shock) $\rightarrow k \times k = 4$
- b: policy $\rightarrow$ output; c: demand $\rightarrow$ rate

Data: the covariance matrix of the surprises

$$\hat{\Sigma} = \begin{pmatrix} 0.246 & 0.010 \\ 0.010 & 0.127 \end{pmatrix}$$

Equations from $\Sigma = \Theta_0 \Theta_0'$:

$$a^2 + b^2 = 0.246 \quad \text{var. of output surprise}$$
$$c^2 + d^2 = 0.127 \quad \text{var. of rate surprise}$$
$$ac + bd = 0.010 \quad \text{covariance}$$

The same regression in matrix form

Two surprises ($u_{y,t}, u_{i,t}$), two shocks (demand $\varepsilon_{d,t}$, policy $\varepsilon_{m,t}$), variance 1

Unknowns:

$$\begin{pmatrix} u_{y,t} \\ u_{i,t} \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\Theta_0} \begin{pmatrix} \varepsilon_{d,t} \\ \varepsilon_{m,t} \end{pmatrix}$$

Data: the covariance matrix of the surprises

$$\hat{\Sigma} = \begin{pmatrix} 0.246 & 0.010 \\ 0.010 & 0.127 \end{pmatrix}$$

- One entry per (variable, shock) $\rightarrow k \times k = 4$
- b: policy $\rightarrow$ output; c: demand $\rightarrow$ rate

Equations from $\Sigma = \Theta_0 \Theta_0'$:

$$a^2 + b^2 = 0.246 \quad \text{var. of output surprise}$$
$$c^2 + d^2 = 0.127 \quad \text{var. of rate surprise}$$
$$ac + bd = 0.010 \quad \text{covariance}$$

- 3 equations, 4 unknowns: we need one more restriction. The regression's choice is $b = 0$:

The same regression in matrix form

Two surprises $(u_{y,t}, u_{i,t})$, two shocks (demand $\varepsilon_{d,t}$, policy $\varepsilon_{m,t}$), variance 1

Unknowns:

$$\begin{pmatrix} u_{y,t} \\ u_{i,t} \end{pmatrix} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\Theta_0} \begin{pmatrix} \varepsilon_{d,t} \\ \varepsilon_{m,t} \end{pmatrix}$$

Data: the covariance matrix of the surprises

$$\hat{\Sigma} = \begin{pmatrix} 0.246 & 0.010 \\ 0.010 & 0.127 \end{pmatrix}$$

Equations from $\Sigma = \Theta_0 \Theta_0'$:

$$a^2 + b^2 = 0.246 \quad \text{var. of output surprise}$$
$$c^2 + d^2 = 0.127 \quad \text{var. of rate surprise}$$
$$ac + bd = 0.010 \quad \text{covariance}$$

- One entry per (variable, shock) $\rightarrow k \times k = 4$
- b: policy $\rightarrow$ output; c: demand $\rightarrow$ rate

- 3 equations, 4 unknowns: we need one more restriction. The regression's choice is $b = 0$:

$\Rightarrow$ $a = \text{sd}(u_y) = 0.50$, $c = \hat{\gamma} \times a = 0.02$, $d = \text{sd}(e) = 0.36$: exactly the regression

One restriction, one answer

One restriction, one answer

Timing: $b = 0$

policy moves output only with a lag

$$\Theta_0 = \begin{pmatrix} 0.50 & 0 \\ 0.02 & 0.36 \end{pmatrix}$$

- Comovement = the Bank reacting to demand
- Policy shock: rate $+0.36$pp, output 0 on impact
- = regress u_i on u_y (recursive / Cholesky)

One restriction, one answer

Timing: $b = 0$

policy moves output only with a lag

$$\Theta_0 = \begin{pmatrix} 0.50 & 0 \\ 0.02 & 0.36 \end{pmatrix}$$

- Comovement = the Bank reacting to demand
- Policy shock: rate +0.36pp, output 0 on impact
- = regress u_i on u_y (recursive / Cholesky)

Opposite: $c = 0$

the Bank ignores this month's output

$$\Theta_0 = \begin{pmatrix} 0.50 & 0.03 \\ 0 & 0.36 \end{pmatrix}$$

- Comovement = policy moving output
- = regress u_y on u_i
- Policy shock: rate +0.36pp, output +0.03%: a tightening raises output?

One restriction, one answer

Timing: $b = 0$

policy moves output only with a lag

$$\Theta_0 = \begin{pmatrix} 0.50 & 0 \\ 0.02 & 0.36 \end{pmatrix}$$

- Comovement = the Bank reacting to demand
- Policy shock: rate +0.36pp, output 0 on impact
- = regress u_i on u_y (recursive / Cholesky)

Opposite: $c = 0$

the Bank ignores this month's output

$$\Theta_0 = \begin{pmatrix} 0.50 & 0.03 \\ 0 & 0.36 \end{pmatrix}$$

- Comovement = policy moving output
- = regress u_y on u_i
- Policy shock: rate +0.36pp, output +0.03%: a tightening raises output?

$\Rightarrow$ Same data, different restriction, different shocks: the restriction is the economics

One restriction, one answer

Timing: $b = 0$

policy moves output only with a lag

$$\Theta_0 = \begin{pmatrix} 0.50 & 0 \\ 0.02 & 0.36 \end{pmatrix}$$

- Comovement = the Bank reacting to demand
- Policy shock: rate +0.36pp, output 0 on impact
- = regress u_i on u_y (recursive / Cholesky)

Opposite: $c = 0$

the Bank ignores this month's output

$$\Theta_0 = \begin{pmatrix} 0.50 & 0.03 \\ 0 & 0.36 \end{pmatrix}$$

- Comovement = policy moving output
- = regress u_y on u_i
- Policy shock: rate +0.36pp, output +0.03%: a tightening raises output?

$\Rightarrow$ Same data, different restriction, different shocks: the restriction is the economics

$\Rightarrow$ UK: the surprises are barely correlated (0.06), so the two nearly agree (V2)

How many restrictions do we need?

How many restrictions do we need?

- The only information on Θ_0 is the covariance matrix of the surprises: $\Sigma = \Theta_0 \Theta_0'$

How many restrictions do we need?

- The only information on Θ_0 is the covariance matrix of the surprises: $\Sigma = \Theta_0 \Theta_0'$
- **Counting equations and unknowns:**
  - Equations: Σ has $k(k+1)/2$ distinct elements $k = 3$: 3 variances + 3 covariances = 6
  - Unknowns: Θ_0 has k^2 entries (each shock on each variable) 9

  $\Rightarrow$ To identify all k shocks we need $k(k-1)/2$ extra restrictions $k = 2$: 1, $k = 3$: 3, $k = 5$: 10

How many restrictions do we need?

- The only information on Θ_0 is the covariance matrix of the surprises: $\Sigma = \Theta_0 \Theta_0'$
- **Counting equations and unknowns:**
  - Equations: Σ has $k(k+1)/2$ distinct elements $k = 3$: 3 variances + 3 covariances = 6
  - Unknowns: Θ_0 has k^2 entries (each shock on each variable) 9

  $\Rightarrow$ To identify all k shocks we need $k(k-1)/2$ extra restrictions $k = 2$: 1, $k = 3$: 3, $k = 5$: 10
- **Recursive ordering** = a chain of regressions, each surprise on those ordered before it
  $\rightarrow$ sets the $k(k-1)/2$ entries above the diagonal of Θ_0 to zero: exactly enough

How many restrictions do we need?

- The only information on Θ_0 is the covariance matrix of the surprises: $\Sigma = \Theta_0 \Theta_0'$
- **Counting equations and unknowns:**
  - Equations: Σ has $k(k+1)/2$ distinct elements — $k = 3$: 3 variances + 3 covariances = 6
  - Unknowns: Θ_0 has k^2 entries (each shock on each variable) — 9

  $\Rightarrow$ To identify all k shocks we need $k(k-1)/2$ extra restrictions $k = 2$: 1, $k = 3$: 3, $k = 5$: 10
- **Recursive ordering** = a chain of regressions, each surprise on those ordered before it
  $\rightarrow$ sets the $k(k-1)/2$ entries above the diagonal of Θ_0 to zero: exactly enough
- **Only one shock** (e.g. policy)? We need only its column of Θ_0: $k-1$ restrictions
  - UK: the policy shock does not move y or u within the month (2 zeros) = rate ordered last
  - Or an instrument for the policy shock (V4)

How many restrictions do we need?

- The only information on Θ_0 is the covariance matrix of the surprises: $\Sigma = \Theta_0 \Theta_0'$
- **Counting equations and unknowns:**
  - Equations: Σ has $k(k+1)/2$ distinct elements — $k = 3$: 3 variances + 3 covariances = 6
  - Unknowns: Θ_0 has k^2 entries (each shock on each variable) — 9

  $\Rightarrow$ To identify all k shocks we need $k(k-1)/2$ extra restrictions $k = 2$: 1, $k = 3$: 3, $k = 5$: 10
- **Recursive ordering** = a chain of regressions, each surprise on those ordered before it
  $\rightarrow$ sets the $k(k-1)/2$ entries above the diagonal of Θ_0 to zero: exactly enough
- **Only one shock** (e.g. policy)? We need only its column of Θ_0: $k-1$ restrictions
  - UK: the policy shock does not move y or u within the month (2 zeros) = rate ordered last
  - Or an instrument for the policy shock (V4)

$\Rightarrow$ **Identification** = supplying the missing restrictions with economics: zeros, an instrument, signs

The same model, written as economics

The same model, written as economics

- Structural form: $A_0 y_t = A_1^* y_{t-1} + \cdots + \varepsilon_t$, with $A_0 = \Theta_0^{-1}$

The same model, written as economics

- Structural form: $A_0 y_t = A_1^* y_{t-1} + \cdots + \varepsilon_t$, with $A_0 = \Theta_0^{-1}$
- Back to the running example, output and the rate only:

$$y_t = \beta\, i_t + \cdots + \omega_y \eta_{y,t}$$
$$i_t = \gamma\, y_t + \cdots + \omega_i \eta_{i,t}$$

The same model, written as economics

- Structural form: $A_0 y_t = A_1^* y_{t-1} + \cdots + \varepsilon_t$, with $A_0 = \Theta_0^{-1}$
- Back to the running example, output and the rate only:

$$y_t = \beta\, i_t + \cdots + \omega_y \eta_{y,t}$$
$$i_t = \gamma\, y_t + \cdots + \omega_i \eta_{i,t}$$

- Unknowns on impact: $\beta, \gamma, \omega_y, \omega_i$ (4). Data: elements of Σ (3) $\Rightarrow$ one short

The same model, written as economics

- Structural form: $A_0 y_t = A_1^* y_{t-1} + \cdots + \varepsilon_t$, with $A_0 = \Theta_0^{-1}$
- Back to the running example, output and the rate only:

$$y_t = \beta\, i_t + \cdots + \omega_y \eta_{y,t}$$
$$i_t = \gamma\, y_t + \cdots + \omega_i \eta_{i,t}$$

- Unknowns on impact: $\beta, \gamma, \omega_y, \omega_i$ (4). Data: elements of Σ (3) $\Rightarrow$ one short
- **One restriction closes the gap:**
  - $\beta = 0$: policy affects output only with a lag $\rightarrow$ recursive, rate ordered last
  - Alternatives: γ known, or a sign on β

Impulse responses

Impulse responses

$$\Theta_h = C_h \Theta_0, \qquad IRF_{ij}(h) = [\Theta_h]_{ij}$$

equivalently: $\Theta_0 = H, \quad \Theta_1 = A_1 \Theta_0, \quad \Theta_2 = A_1 \Theta_1 + A_2 \Theta_0, \quad \ldots, \quad \Theta_h = \sum_{\ell=1}^{\min(p,h)} A_\ell \, \Theta_{h-\ell}$

Impulse responses

$$\Theta_h = C_h \Theta_0, \qquad IRF_{ij}(h) = [\Theta_h]_{ij}$$

equivalently: $\Theta_0 = H, \quad \Theta_1 = A_1 \Theta_0, \quad \Theta_2 = A_1 \Theta_1 + A_2 \Theta_0, \quad \ldots, \quad \Theta_h = \sum_{\ell=1}^{\min(p,h)} A_\ell \, \Theta_{h-\ell}$

- Reduced-form dynamics $\times$ impact matrix: every horizon comes from the one-step model

Impulse responses

$$\Theta_h = C_h \Theta_0, \qquad IRF_{ij}(h) = [\Theta_h]_{ij}$$

equivalently: $\Theta_0 = H, \quad \Theta_1 = A_1 \Theta_0, \quad \Theta_2 = A_1 \Theta_1 + A_2 \Theta_0, \quad \ldots, \quad \Theta_h = \sum_{\ell=1}^{\min(p,h)} A_\ell \, \Theta_{h-\ell}$

- Reduced-form dynamics $\times$ impact matrix: every horizon comes from the one-step model
- **Scale:**
  - One-standard-deviation shock: a column of Θ_0
  - Unit effect: divide by impact on the policy variable $\rightarrow$ "+1pp on the rate"

Impulse responses

$$\Theta_h = C_h \Theta_0, \qquad IRF_{ij}(h) = [\Theta_h]_{ij}$$

equivalently: $\Theta_0 = H, \quad \Theta_1 = A_1 \Theta_0, \quad \Theta_2 = A_1 \Theta_1 + A_2 \Theta_0, \quad \ldots, \quad \Theta_h = \sum_{\ell=1}^{\min(p,h)} A_\ell \, \Theta_{h-\ell}$

- Reduced-form dynamics $\times$ impact matrix: every horizon comes from the one-step model
- **Scale:**
  - One-standard-deviation shock: a column of Θ_0
  - Unit effect: divide by impact on the policy variable $\rightarrow$ "+1pp on the rate"
- **Cumulate** if a variable enters in growth rates

Impulse responses

$$\Theta_h = C_h \Theta_0, \qquad IRF_{ij}(h) = [\Theta_h]_{ij}$$

equivalently: $\Theta_0 = H, \quad \Theta_1 = A_1 \Theta_0, \quad \Theta_2 = A_1 \Theta_1 + A_2 \Theta_0, \quad \ldots, \quad \Theta_h = \sum_{\ell=1}^{\min(p,h)} A_\ell \, \Theta_{h-\ell}$

- Reduced-form dynamics $\times$ impact matrix: every horizon comes from the one-step model
- **Scale:**
  - One-standard-deviation shock: a column of Θ_0
  - Unit effect: divide by impact on the policy variable $\rightarrow$ "+1pp on the rate"
- **Cumulate** if a variable enters in growth rates
- **Partial:** only the column of shock j is needed, $\Theta_h e_j = C_h \theta_j$

Recursive (Cholesky) identification

Recursive (Cholesky) identification

- $\Theta_0 = \text{chol}(\Sigma)$, lower triangular
  - Variable 1 responds on impact only to shock 1, variable 2 to shocks 1 and 2, ...

  $\Rightarrow$ The ordering is the assumption

Recursive (Cholesky) identification

- $\Theta_0 = \text{chol}(\Sigma)$, lower triangular
  - Variable 1 responds on impact only to shock 1, variable 2 to shocks 1 and 2, ...

  $\Rightarrow$ The ordering is the assumption
- **Monetary policy** [Christiano, Eichenbaum & Evans (1999)]
  - Slow block (output, prices) $\rightarrow$ policy rate $\rightarrow$ fast block (money, asset prices)
  - Policy reacts to today's macro; macro reacts to policy with a lag

Recursive (Cholesky) identification

- $\Theta_0 = \text{chol}(\Sigma)$, lower triangular
  - Variable 1 responds on impact only to shock 1, variable 2 to shocks 1 and 2, ...

  $\Rightarrow$ The ordering is the assumption
- **Monetary policy** [Christiano, Eichenbaum & Evans (1999)]
  - Slow block (output, prices) $\rightarrow$ policy rate $\rightarrow$ fast block (money, asset prices)
  - Policy reacts to today's macro; macro reacts to policy with a lag
- **Block recursion:** to identify only the policy shock, the ordering *within* blocks does not matter

Recursive (Cholesky) identification

- $\Theta_0 = \text{chol}(\Sigma)$, lower triangular
  - Variable 1 responds on impact only to shock 1, variable 2 to shocks 1 and 2, ...

  $\Rightarrow$ The ordering is the assumption
- **Monetary policy** [Christiano, Eichenbaum & Evans (1999)]
  - Slow block (output, prices) $\rightarrow$ policy rate $\rightarrow$ fast block (money, asset prices)
  - Policy reacts to today's macro; macro reacts to policy with a lag
- **Block recursion:** to identify only the policy shock, the ordering *within* blocks does not matter
- Credible for slow real variables at monthly frequency
  Not credible for asset prices and policy: both react within the day

Recursive (Cholesky) identification

- $\Theta_0 = \text{chol}(\Sigma)$, lower triangular
  - Variable 1 responds on impact only to shock 1, variable 2 to shocks 1 and 2, ...

  $\Rightarrow$ The ordering is the assumption
- **Monetary policy** [Christiano, Eichenbaum & Evans (1999)]
  - Slow block (output, prices) $\rightarrow$ policy rate $\rightarrow$ fast block (money, asset prices)
  - Policy reacts to today's macro; macro reacts to policy with a lag
- **Block recursion:** to identify only the policy shock, the ordering *within* blocks does not matter
- Credible for slow real variables at monthly frequency
  Not credible for asset prices and policy: both react within the day

$\Rightarrow$ Same assumption as the LP with y_t, u_t as controls (L2): with the same p, both use exactly the same shock [Plagborg-Møller & Wolf (2021)]

Inference: how sure are we about the response?

Inference: how sure are we about the response?

- Our impulse response is estimated from one sample (356 months)
  $\rightarrow$ another sample from the same economy would give a different line

Inference: how sure are we about the response?

- Our impulse response is estimated from one sample (356 months)
  $\rightarrow$ another sample from the same economy would give a different line
- **Inference** = quantify that sampling uncertainty
  - Confidence bands around $\hat{\Theta}_h$
  - Tests: "is output lower after 2 years?" $\rightarrow$ does the band at $h = 24$ exclude zero?

Inference: how sure are we about the response?

- Our impulse response is estimated from one sample (356 months)
  $\rightarrow$ another sample from the same economy would give a different line
- **Inference** = quantify that sampling uncertainty
  - Confidence bands around $\hat{\Theta}_h$
  - Tests: "is output lower after 2 years?" $\rightarrow$ does the band at $h = 24$ exclude zero?
- **What to choose and report:**
  1. How to compute the bands (next slide)
  2. Coverage: 68% and 90% are the conventions
  3. Pointwise (each h separately) or for the whole path (sup-t band)

Inference: how sure are we about the response?

- Our impulse response is estimated from one sample (356 months)
  $\rightarrow$ another sample from the same economy would give a different line
- **Inference** = quantify that sampling uncertainty
  - Confidence bands around $\hat{\Theta}_h$
  - Tests: "is output lower after 2 years?" $\rightarrow$ does the band at $h = 24$ exclude zero?
- **What to choose and report:**
  1. How to compute the bands (next slide)
  2. Coverage: 68% and 90% are the conventions
  3. Pointwise (each h separately) or for the whole path (sup-t band)

$\Rightarrow$ Bands do not cover bias (too few lags) or a wrong identifying assumption

Three ways to compute VAR bands

Three ways to compute VAR bands

- **Delta method** (formula): unlike in an LP, Θ_h is not a regression coefficient
  - OLS gives SEs for $\hat{A}$ only; translate them to $\Theta_h = g(\hat{A})$ via the slope of g: $\text{Var}(\hat{\Theta}_h) \approx g' \,\text{Var}(\hat{A})\, g'^{\top} \rightarrow$ band $\hat{\Theta}_h \pm 1.645 \cdot SE$
  - Poor at long horizons: Θ_h involves A^h, far from linear when the data are persistent

Three ways to compute VAR bands

- **Delta method** (formula): unlike in an LP, Θ_h is not a regression coefficient
  - OLS gives SEs for $\hat{A}$ only; translate them to $\Theta_h = g(\hat{A})$ via the slope of g: $\text{Var}(\hat{\Theta}_h) \approx g' \,\text{Var}(\hat{A})\, g'^{\top} \rightarrow$ band $\hat{\Theta}_h \pm 1.645 \cdot SE$
  - Poor at long horizons: Θ_h involves A^h, far from linear when the data are persistent
- **Residual bootstrap** (simulation; used in V1): "rerun history with different luck" [Kilian (1998)]
  1. Draw 344 residual vectors from the estimated $\hat{u}_t$ with replacement (each can be drawn again)
  2. Feed them through the estimated VAR $\rightarrow$ an artificial UK history y_t^*
  3. Re-estimate the VAR and the responses on y^*; repeat 500 times
  4. 90% band = 5th to 95th percentile of the 500 responses

Three ways to compute VAR bands

- **Delta method** (formula): unlike in an LP, Θ_h is not a regression coefficient
  - OLS gives SEs for $\hat{A}$ only; translate them to $\Theta_h = g(\hat{A})$ via the slope of g: $\text{Var}(\hat{\Theta}_h) \approx g' \,\text{Var}(\hat{A})\, g'^{\top} \rightarrow$ band $\hat{\Theta}_h \pm 1.645 \cdot SE$
  - Poor at long horizons: Θ_h involves A^h, far from linear when the data are persistent
- **Residual bootstrap** (simulation; used in V1): "rerun history with different luck" [Kilian (1998)]
  1. Draw 344 residual vectors from the estimated $\hat{u}_t$ with replacement (each can be drawn again)
  2. Feed them through the estimated VAR $\rightarrow$ an artificial UK history y_t^*
  3. Re-estimate the VAR and the responses on y^*; repeat 500 times
  4. 90% band = 5th to 95th percentile of the 500 responses
- **Bayesian** (beliefs): treat A and Σ as random
  - Prior: each series close to a random walk, distant lags close to zero (Minnesota prior)
  - Prior + data $\rightarrow$ posterior; draw (A, Σ), compute responses, take percentiles [Giannone, Lenza & Primiceri (2015)]

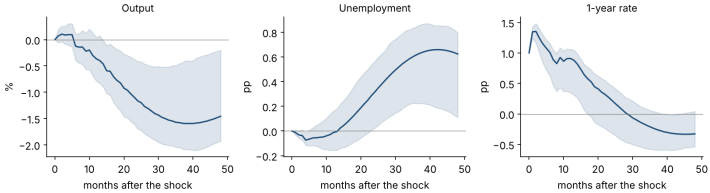
V1: a recursive VAR

$$x_t = c + \sum_{\ell=1}^{12} A_\ell \, x_{t-\ell} + u_t, \quad x_t = (y_t, u_t, i_t)', \qquad u_t = \Theta_0 \, \varepsilon_t, \;\; \Theta_0 = \text{chol}(\Sigma), \qquad \Theta_h = \sum_{\ell=1}^{\min(h,12)} A_\ell \, \Theta_{h-\ell}$$



V1: a recursive VAR

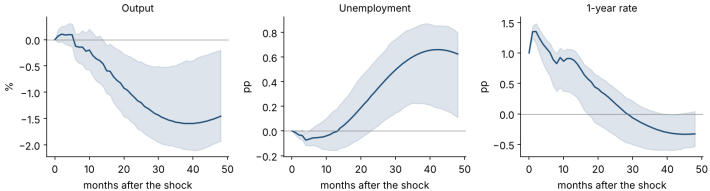
$$x_t = c + \sum_{\ell=1}^{12} A_\ell\, x_{t-\ell} + u_t, \quad x_t = (y_t, u_t, i_t)', \qquad u_t = \Theta_0\, \varepsilon_t, \;\; \Theta_0 = \text{chol}(\Sigma), \qquad \Theta_h = \sum_{\ell=1}^{\min(h,12)} A_\ell\, \Theta_{h-\ell}$$



- Policy shock: third column of Θ_0, scaled to +1pp on i; 90% residual-bootstrap bands

V1: a recursive VAR

$$x_t = c + \sum_{\ell=1}^{12} A_\ell\, x_{t-\ell} + u_t, \quad x_t = (y_t, u_t, i_t)', \qquad u_t = \Theta_0\, \varepsilon_t, \;\; \Theta_0 = \mathrm{chol}(\Sigma), \qquad \Theta_h = \sum_{\ell=1}^{\min(h,12)} A_\ell\, \Theta_{h-\ell}$$



- Policy shock: third column of Θ_0, scaled to +1pp on i; 90% residual-bootstrap bands
- Output -0.36% at 12 months, -1.16% at 24, -1.58% at 36; unemployment up to $+0.66$pp

V1: a recursive VAR

$$x_t = c + \sum_{\ell=1}^{12} A_\ell\, x_{t-\ell} + u_t, \quad x_t = (y_t, u_t, i_t)', \qquad u_t = \Theta_0\, \varepsilon_t, \;\; \Theta_0 = \text{chol}(\Sigma), \qquad \Theta_h = \sum_{\ell=1}^{\min(h,12)} A_\ell\, \Theta_{h-\ell}$$

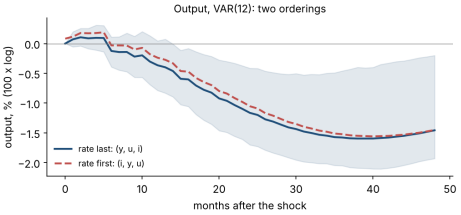


- Policy shock: third column of Θ_0, scaled to +1pp on i; 90% residual-bootstrap bands
- Output -0.36% at 12 months, -1.16% at 24, -1.58% at 36; unemployment up to $+0.66$pp

$\Rightarrow$ Assumption: output and unemployment do not react to the policy shock within the month

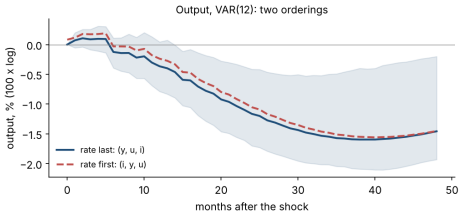
V2: the ordering is the assumption

rate last: $\Theta_0 = \text{chol}(\Sigma_{(y,u,i)})$, 3rd column vs rate first: $\Theta_0 = \text{chol}(\Sigma_{(i,y,u)})$, 1st column



V2: the ordering is the assumption

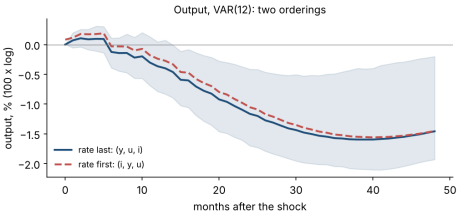
rate last: $\Theta_0 = \text{chol}(\Sigma_{(y,u,i)})$, 3rd column vs rate first: $\Theta_0 = \text{chol}(\Sigma_{(i,y,u)})$, 1st column



- Rate first: the shock is all of $u_{i,t}$, and output may move on impact ($+0.08$); -1.05% at 24 months

V2: the ordering is the assumption

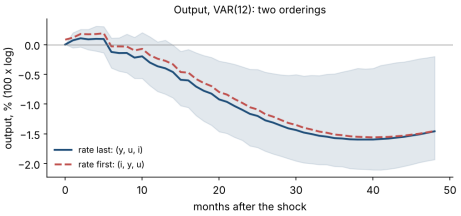
rate last: $\Theta_0 = \text{chol}(\Sigma_{(y,u,i)})$, 3rd column vs rate first: $\Theta_0 = \text{chol}(\Sigma_{(i,y,u)})$, 1st column



- Rate first: the shock is all of $u_{i,t}$, and output may move on impact ($+0.08$); -1.05% at 24 months
- The ordering matters only as much as the forecast errors are correlated: here $\text{corr}(u_{y,t}, u_{i,t}) = 0.06$

V2: the ordering is the assumption

rate last: $\Theta_0 = \text{chol}(\Sigma_{(y,u,i)})$, 3rd column vs rate first: $\Theta_0 = \text{chol}(\Sigma_{(i,y,u)})$, 1st column

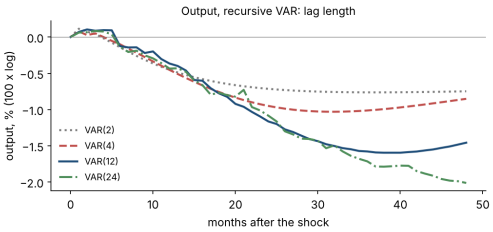


- Rate first: the shock is all of $u_{i,t}$, and output may move on impact ($+0.08$); -1.05% at 24 months
- The ordering matters only as much as the forecast errors are correlated: here $\text{corr}(u_{y,t}, u_{i,t}) = 0.06$

$\Rightarrow$ With monthly data and few variables, timing assumptions bite less than people fear; with quarterly data, much more

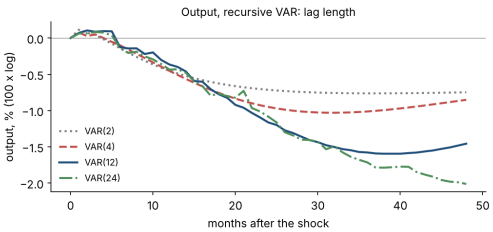
V3: the lag length

$$x_t = c + \sum_{\ell=1}^{p} A_\ell \, x_{t-\ell} + u_t, \qquad p \in \{2, 4, 12, 24\}, \qquad \text{recursive } (y, u, i)$$



V3: the lag length

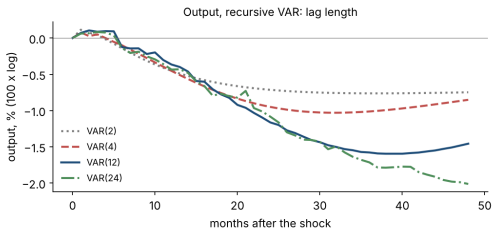
$$x_t = c + \sum_{\ell=1}^{p} A_\ell \, x_{t-\ell} + u_t, \qquad p \in \{2, 4, 12, 24\}, \qquad \text{recursive } (y, u, i)$$



- Close for the first year; at 48 months: -0.75 ($p = 2$), -0.85 (4), -1.46 (12), -2.01 (24)

V3: the lag length

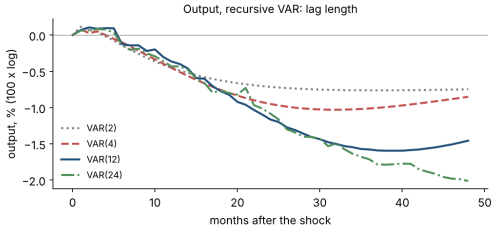
$$x_t = c + \sum_{\ell=1}^{p} A_\ell \, x_{t-\ell} + u_t, \qquad p \in \{2, 4, 12, 24\}, \qquad \text{recursive } (y, u, i)$$



- Close for the first year; at 48 months: -0.75 ($p = 2$), -0.85 (4), -1.46 (12), -2.01 (24)
- AIC picks $p = 5$, BIC $p = 3$: the long-run answer depends on a choice the data barely pin down

V3: the lag length

$$x_t = c + \sum_{\ell=1}^{p} A_\ell \, x_{t-\ell} + u_t, \qquad p \in \{2, 4, 12, 24\}, \qquad \text{recursive } (y, u, i)$$



- Close for the first year; at 48 months: -0.75 ($p = 2$), -0.85 (4), -1.46 (12), -2.01 (24)
- AIC picks $p = 5$, BIC $p = 3$: the long-run answer depends on a choice the data barely pin down

$\Rightarrow$ Beyond $h = p$ the VAR extrapolates; VAR(24) finds a slow wave (trough at 4 years, back near 0 by 10)

Proxy SVAR: 2SLS on the VAR surprises

Proxy SVAR: 2SLS on the VAR surprises

- Two variables, output and the rate: the VAR gives surprises $u_{y,t}$, $u_{i,t}$
  - Recursive: regress u_i on u_y (timing). **Instead:** use the narrative shock z_t as an instrument

Proxy SVAR: 2SLS on the VAR surprises

- Two variables, output and the rate: the VAR gives surprises $u_{y,t}$, $u_{i,t}$
  - Recursive: regress u_i on u_y (timing). **Instead:** use the narrative shock z_t as an instrument
- **2SLS on the surprises:** $u_{y,t} = \beta\, u_{i,t} + e_t$, $u_{i,t}$ instrumented by z_t
  - First stage: $u_{i,t}$ on z_t: coefficient 0.36, robust $F = 31 \rightarrow$ relevant
  - $\hat{\beta} = \frac{\mathrm{Cov}(u_y, z)}{\mathrm{Cov}(u_i, z)} = \frac{0.015}{0.044} = 0.34$: impact effect on output of a policy-driven 1pp rate rise

Proxy SVAR: 2SLS on the VAR surprises

- Two variables, output and the rate: the VAR gives surprises $u_{y,t}$, $u_{i,t}$
  - Recursive: regress u_i on u_y (timing). **Instead:** use the narrative shock z_t as an instrument
- **2SLS on the surprises:** $u_{y,t} = \beta\, u_{i,t} + e_t$, $u_{i,t}$ instrumented by z_t
  - First stage: $u_{i,t}$ on z_t: coefficient 0.36, robust $F = 31 \rightarrow$ relevant
  - $\hat{\beta} = \frac{\text{Cov}(u_y, z)}{\text{Cov}(u_i, z)} = \frac{0.015}{0.044} = 0.34$: impact effect on output of a policy-driven 1pp rate rise
- **Why surprises, not y_t and i_t?** The VAR's lags already removed the predictable part (like LP controls)

Proxy SVAR: 2SLS on the VAR surprises

- Two variables, output and the rate: the VAR gives surprises $u_{y,t}$, $u_{i,t}$
  - Recursive: regress u_i on u_y (timing). **Instead:** use the narrative shock z_t as an instrument
- **2SLS on the surprises:** $u_{y,t} = \beta\, u_{i,t} + e_t$, $u_{i,t}$ instrumented by z_t
  - First stage: $u_{i,t}$ on z_t: coefficient 0.36, robust $F = 31 \rightarrow$ relevant
  - $\hat{\beta} = \frac{\mathrm{Cov}(u_y, z)}{\mathrm{Cov}(u_i, z)} = \frac{0.015}{0.044} = 0.34$: impact effect on output of a policy-driven 1pp rate rise
- **Why surprises, not y_t and i_t?** The VAR's lags already removed the predictable part (like LP controls)
- **Horizons $h \geq 1$:** the VAR's dynamics carry the impact effect forward, $\Theta_h \theta = C_h \theta$

Proxy SVAR: 2SLS on the VAR surprises

- Two variables, output and the rate: the VAR gives surprises $u_{y,t}$, $u_{i,t}$
  - Recursive: regress u_i on u_y (timing). **Instead:** use the narrative shock z_t as an instrument
- **2SLS on the surprises:** $u_{y,t} = \beta\, u_{i,t} + e_t$, $u_{i,t}$ instrumented by z_t
  - First stage: $u_{i,t}$ on z_t: coefficient 0.36, robust $F = 31 \rightarrow$ relevant
  - $\hat{\beta} = \frac{\mathrm{Cov}(u_y, z)}{\mathrm{Cov}(u_i, z)} = \frac{0.015}{0.044} = 0.34$: impact effect on output of a policy-driven 1pp rate rise
- **Why surprises, not y_t and i_t?** The VAR's lags already removed the predictable part (like LP controls)
- **Horizons $h \geq 1$:** the VAR's dynamics carry the impact effect forward, $\Theta_h\theta = C_h\theta$

$\Rightarrow$ **LP-IV:** 2SLS at every horizon. **SVAR-IV:** 2SLS only on impact, the VAR does the rest

Proxy SVAR: 2SLS on the VAR surprises

- Two variables, output and the rate: the VAR gives surprises $u_{y,t}$, $u_{i,t}$
  - Recursive: regress u_i on u_y (timing). **Instead:** use the narrative shock z_t as an instrument
- **2SLS on the surprises:** $u_{y,t} = \beta\, u_{i,t} + e_t$, $u_{i,t}$ instrumented by z_t
  - First stage: $u_{i,t}$ on z_t: coefficient 0.36, robust $F = 31 \rightarrow$ relevant
  - $\hat{\beta} = \frac{\mathrm{Cov}(u_y, z)}{\mathrm{Cov}(u_i, z)} = \frac{0.015}{0.044} = 0.34$: impact effect on output of a policy-driven 1pp rate rise
- **Why surprises, not y_t and i_t?** The VAR's lags already removed the predictable part (like LP controls)
- **Horizons $h \geq 1$:** the VAR's dynamics carry the impact effect forward, $\Theta_h \theta = C_h \theta$

$\Rightarrow$ **LP-IV:** 2SLS at every horizon. **SVAR-IV:** 2SLS only on impact, the VAR does the rest

$\Rightarrow$ UK: output rises on impact ($+0.34$): a warning that z is not entirely clean (V4)

Two ways to use the instrument in a VAR

Two ways to use the instrument in a VAR

1) External instrument (SVAR-IV)

the baseline

- VAR in (y, u, i); z_t stays outside
- 2SLS of the surprises on $z_t \rightarrow$ impact effects
- The VAR's dynamics carry them forward
- Needs invertibility: the shock must show up in this month's surprises

Two ways to use the instrument in a VAR

1) External instrument (SVAR-IV)

the baseline

- VAR in (y, u, i); z_t stays outside
- 2SLS of the surprises on $z_t \rightarrow$ impact effects
- The VAR's dynamics carry them forward
- Needs invertibility: the shock must show up in this month's surprises

2) Internal instrument

often more useful

- Add z_t as a fourth series: VAR in (z, y, u, i), recursive, z first
- Shock = surprise in z_t, scaled to $+1$pp on the rate
- y, u, i may all react within the month
- No invertibility needed; lags clean any predictable part of z_t

Two ways to use the instrument in a VAR

1) External instrument (SVAR-IV)
the baseline

- VAR in (y, u, i); z_t stays outside
- 2SLS of the surprises on $z_t \rightarrow$ impact effects
- The VAR's dynamics carry them forward
- Needs invertibility: the shock must show up in this month's surprises

2) Internal instrument
often more useful

- Add z_t as a fourth series: VAR in (z, y, u, i), recursive, z first
- Shock = surprise in z_t, scaled to $+1$pp on the rate
- y, u, i may all react within the month
- No invertibility needed; lags clean any predictable part of z_t

$\Rightarrow$ Same instrument and exogeneity assumption; internal targets the same object as LP-IV (L3)
[Plagborg-Møller & Wolf (2021)]

Two ways to use the instrument in a VAR

1) External instrument (SVAR-IV)
the baseline

- VAR in (y, u, i); z_t stays outside
- 2SLS of the surprises on $z_t \rightarrow$ impact effects
- The VAR's dynamics carry them forward
- Needs invertibility: the shock must show up in this month's surprises

2) Internal instrument
often more useful

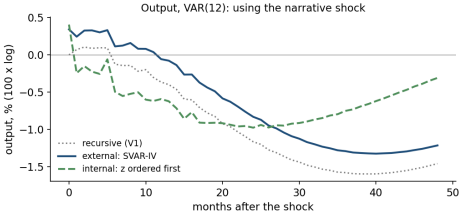
- Add z_t as a fourth series: VAR in (z, y, u, i), recursive, z first
- Shock = surprise in z_t, scaled to $+1$pp on the rate
- y, u, i may all react within the month
- No invertibility needed; lags clean any predictable part of z_t

$\Rightarrow$ Same instrument and exogeneity assumption; internal targets the same object as LP-IV (L3) [Plagborg-Møller & Wolf (2021)]

$\Rightarrow$ If both are valid they agree; if they disagree, trust the internal one (V4)

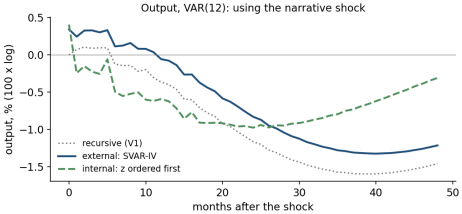
V4: the narrative shock in a VAR

1) External: $\hat{u}_{j,t} = b_j\, z_t + e_{j,t}, \;\; \theta = b/b_i, \;\; \Theta_h\theta = C_h\theta$ **2) Internal:** $(z_t, y_t, u_t, i_t)'$ recursive, z first



V4: the narrative shock in a VAR

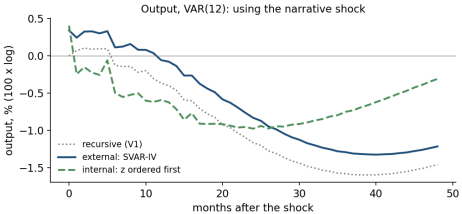
1) External: $\hat{u}_{j,t} = b_j\, z_t + e_{j,t}$, $\theta = b/b_i$, $\Theta_h\theta = C_h\theta$ **2) Internal:** $(z_t, y_t, u_t, i_t)'$ recursive, z first



- Output at 12 / 24 / 36 months: external -0.06 / -0.83 / -1.29; internal -0.59 / -0.98 / -0.75

V4: the narrative shock in a VAR

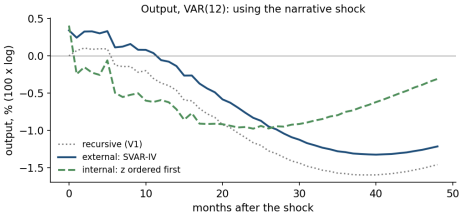
1) External: $\hat{u}_{j,t} = b_j\, z_t + e_{j,t}$, $\theta = b/b_i$, $\Theta_h\theta = C_h\theta$ **2) Internal:** $(z_t, y_t, u_t, i_t)'$ recursive, z first



- Output at 12 / 24 / 36 months: external -0.06 / -0.83 / -1.29; internal -0.59 / -0.98 / -0.75
- External starts from output rising on impact ($+0.34$) and the VAR propagates it; internal matches LP-IV (L3)

V4: the narrative shock in a VAR

1) External: $\hat{u}_{j,t} = b_j\, z_t + e_{j,t}$, $\theta = b/b_i$, $\Theta_h\theta = C_h\theta$ **2) Internal:** $(z_t, y_t, u_t, i_t)'$ recursive, z first



- Output at 12 / 24 / 36 months: external -0.06 / -0.83 / -1.29; internal -0.59 / -0.98 / -0.75
- External starts from output rising on impact ($+0.34$) and the VAR propagates it; internal matches LP-IV (L3)

$\Rightarrow$ All agree on about -1% at 2 years; trust the internal instrument for the shape

Variance and historical decompositions

Variance and historical decompositions

- **FEVD:** share of h-step forecast error variance of i due to shock j

$$FEVD_{ij}(h) = \frac{\sum_{\ell=0}^{h} [\Theta_\ell]_{ij}^2}{\sum_{\ell=0}^{h} \sum_m [\Theta_\ell]_{im}^2}$$

Variance and historical decompositions

- **FEVD:** share of h-step forecast error variance of i due to shock j

$$FEVD_{ij}(h) = \frac{\sum_{\ell=0}^{h} [\Theta_\ell]_{ij}^2}{\sum_{\ell=0}^{h} \sum_m [\Theta_\ell]_{im}^2}$$

- **Historical decomposition:** contribution of each shock to the path of y_t

$$y_t - \mathbb{E}_{t_0} y_t = \sum_j \sum_\ell [\Theta_\ell]_{\cdot j} \, \varepsilon_{j,t-\ell}$$

Variance and historical decompositions

- **FEVD:** share of h-step forecast error variance of i due to shock j

$$FEVD_{ij}(h) = \frac{\sum_{\ell=0}^{h} [\Theta_\ell]_{ij}^2}{\sum_{\ell=0}^{h} \sum_m [\Theta_\ell]_{im}^2}$$

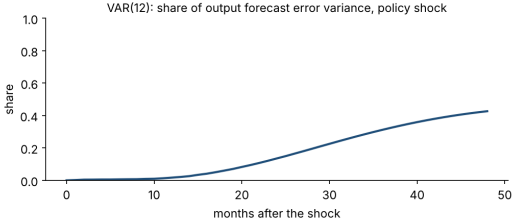
- **Historical decomposition:** contribution of each shock to the path of y_t

$$y_t - \mathbb{E}_{t_0} y_t = \sum_j \sum_\ell [\Theta_\ell]_{\cdot j} \, \varepsilon_{j,t-\ell}$$

$\Rightarrow$ Large, significant responses can still be quantitatively minor

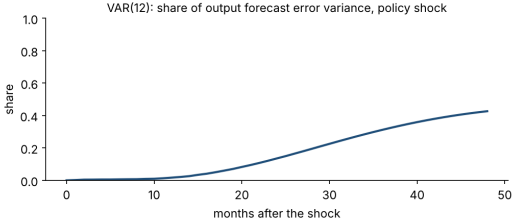
V5: how important is the policy shock?

$$FEVD_y(h) = \frac{\sum_{\ell=0}^{h-1} [\Theta_\ell]_{y,i}^2}{\sum_{\ell=0}^{h-1} \sum_{m \in \{y,u,i\}} [\Theta_\ell]_{y,m}^2}, \qquad \Theta_0 = \text{chol}(\Sigma) \ \text{(one-s.d. shocks)}$$



V5: how important is the policy shock?

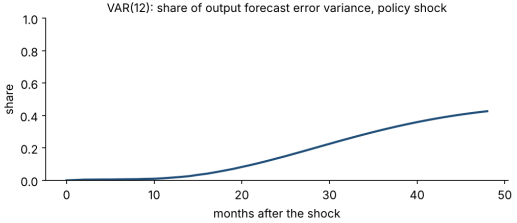
$$FEVD_y(h) = \frac{\sum_{\ell=0}^{h-1} [\Theta_\ell]_{y,i}^2}{\sum_{\ell=0}^{h-1} \sum_{m \in \{y,u,i\}} [\Theta_\ell]_{y,m}^2}, \qquad \Theta_0 = \text{chol}(\Sigma) \ \text{(one-s.d. shocks)}$$



- Policy shocks explain 1% of the output forecast error variance at 12 months, 12% at 24, 42% at 48

V5: how important is the policy shock?

$$FEVD_y(h) = \frac{\sum_{\ell=0}^{h-1} [\Theta_\ell]^2_{y,i}}{\sum_{\ell=0}^{h-1} \sum_{m \in \{y,u,i\}} [\Theta_\ell]^2_{y,m}}, \qquad \Theta_0 = \text{chol}(\Sigma) \ \text{(one-s.d. shocks)}$$



- Policy shocks explain 1% of the output forecast error variance at 12 months, 12% at 24, 42% at 48

$\Rightarrow$ Unimportant in the short run, a major driver at business-cycle horizons (if the VAR's extrapolation is right)

A map of identification schemes

A map of identification schemes

Scheme	Assumption	Reference	Result
Short-run zeros	No response on impact	Christiano et al. (1999)	Point
Long-run zeros	No permanent effect	Blanchard & Quah (1989)	Point
Sign restrictions	Signs for a few periods	Uhlig (2005)	Set
External instruments	Proxy correlated only with target shock	Mertens & Ravn (2013)	Point
Narrative	Dated episodes of large shocks	Romer & Romer (2004)	Point/set
Heteroskedasticity	Shock variances change across regimes	Rigobon (2003)	Point
Non-Gaussianity	Independent, non-Gaussian shocks	Lanne et al. (2017)	Point

A map of identification schemes

Scheme	Assumption	Reference	Result
Short-run zeros	No response on impact	Christiano et al. (1999)	Point
Long-run zeros	No permanent effect	Blanchard & Quah (1989)	Point
Sign restrictions	Signs for a few periods	Uhlig (2005)	Set
External instruments	Proxy correlated only with target shock	Mertens & Ravn (2013)	Point
Narrative	Dated episodes of large shocks	Romer & Romer (2004)	Point/set
Heteroskedasticity	Shock variances change across regimes	Rigobon (2003)	Point
Non-Gaussianity	Independent, non-Gaussian shocks	Lanne et al. (2017)	Point

- Today in detail: recursive, sign restrictions, external instruments

Other schemes in one slide

Other schemes in one slide

- **Sign restrictions:** keep every Θ_0 that fits Σ and gives the expected signs (tightening: rate $\uparrow$, inflation $\downarrow$)
  $\rightarrow$ a set of responses, not one; output often ambiguous [Uhlig (2005); Baumeister & Hamilton (2015)]

Other schemes in one slide

- **Sign restrictions:** keep every Θ_0 that fits Σ and gives the expected signs (tightening: rate $\uparrow$, inflation $\downarrow$)
  $\rightarrow$ a set of responses, not one; output often ambiguous [Uhlig (2005); Baumeister & Hamilton (2015)]

- **Long-run restrictions:** demand shocks have no permanent effect on output
  $\rightarrow$ $C(1)\Theta_0$ triangular; sensitive to low-frequency misspecification [Faust & Leeper (1997)]

Other schemes in one slide

- **Sign restrictions:** keep every Θ_0 that fits Σ and gives the expected signs (tightening: rate $\uparrow$, inflation $\downarrow$)
  $\rightarrow$ a set of responses, not one; output often ambiguous [Uhlig (2005); Baumeister & Hamilton (2015)]

- **Long-run restrictions:** demand shocks have no permanent effect on output
  $\rightarrow$ $C(1)\Theta_0$ triangular; sensitive to low-frequency misspecification [Faust & Leeper (1997)]

- **Heteroskedasticity:** variances shift across regimes, Θ_0 does not [Rigobon (2003)]

Other schemes in one slide

- **Sign restrictions:** keep every Θ_0 that fits Σ and gives the expected signs (tightening: rate $\uparrow$, inflation $\downarrow$)
  $\rightarrow$ a set of responses, not one; output often ambiguous [Uhlig (2005); Baumeister & Hamilton (2015)]

- **Long-run restrictions:** demand shocks have no permanent effect on output
  $\rightarrow$ $C(1)\Theta_0$ triangular; sensitive to low-frequency misspecification [Faust & Leeper (1997)]

- **Heteroskedasticity:** variances shift across regimes, Θ_0 does not [Rigobon (2003)]

- **Non-Gaussianity:** independent non-Gaussian shocks identify Θ_0 up to sign and order
  $\rightarrow$ statistical identification: you still have to name the shocks

Other schemes in one slide

- **Sign restrictions:** keep every Θ_0 that fits Σ and gives the expected signs (tightening: rate $\uparrow$, inflation $\downarrow$)
  $\rightarrow$ a set of responses, not one; output often ambiguous [Uhlig (2005); Baumeister & Hamilton (2015)]

- **Long-run restrictions:** demand shocks have no permanent effect on output
  $\rightarrow$ $C(1)\Theta_0$ triangular; sensitive to low-frequency misspecification [Faust & Leeper (1997)]

- **Heteroskedasticity:** variances shift across regimes, Θ_0 does not [Rigobon (2003)]

- **Non-Gaussianity:** independent non-Gaussian shocks identify Θ_0 up to sign and order
  $\rightarrow$ statistical identification: you still have to name the shocks

$\Rightarrow$ If you can measure a shock directly (even with noise), do it: more transparent and more robust to nonlinearities [Kolesár & Plagborg-Møller (2025)]

Comparing LPs and VARs

Same estimand

Same estimand

- **Result** [Plagborg-Møller & Wolf (2021)]: in population, an LP with p lags and a VAR(p) with the same identification give identical responses for $h \leq p$
  - With $p = \infty$: identical at every horizon

Same estimand

- **Result** [Plagborg-Møller & Wolf (2021)]: in population, an LP with p lags and a VAR(p) with the same identification give identical responses for $h \leq p$
  - With $p = \infty$: identical at every horizon
- **Intuition:**
  - Both project on the same shock: $\tilde{x}_t$ = residual of $u_{x,t}$ on $u_{r,t}$ = residual of x_t on r_t and p lags
  - LP uses autocovariances up to lag h directly
  - VAR matches the first p autocovariances and extrapolates beyond

Same estimand

- **Result** [Plagborg-Møller & Wolf (2021)]: in population, an LP with p lags and a VAR(p) with the same identification give identical responses for $h \leq p$
  - With $p = \infty$: identical at every horizon
- **Intuition:**
  - Both project on the same shock: $\tilde{x}_t$ = residual of $u_{x,t}$ on $u_{r,t}$ = residual of x_t on r_t and p lags
  - LP uses autocovariances up to lag h directly
  - VAR matches the first p autocovariances and extrapolates beyond
- Holds for recursive, IV, long-run and sign identification

Same estimand

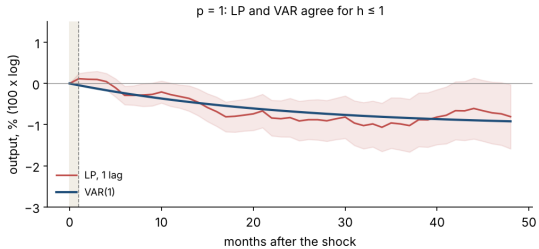
- **Result** [Plagborg-Møller & Wolf (2021)]: in population, an LP with p lags and a VAR(p) with the same identification give identical responses for $h \leq p$
  - With $p = \infty$: identical at every horizon
- **Intuition:**
  - Both project on the same shock: $\tilde{x}_t$ = residual of $u_{x,t}$ on $u_{r,t}$ = residual of x_t on r_t and p lags
  - LP uses autocovariances up to lag h directly
  - VAR matches the first p autocovariances and extrapolates beyond
- Holds for recursive, IV, long-run and sign identification

$\Rightarrow$ Anything you can identify with an SVAR you can identify with an LP, and vice versa

$\Rightarrow$ The choice is about bias and variance in finite samples

C1: same shock, horizon by horizon

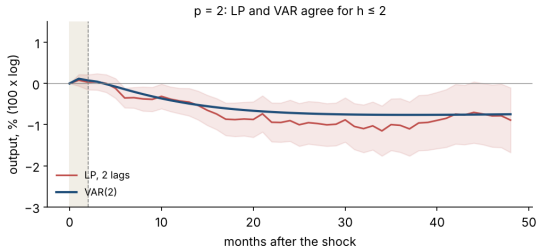
LP: $y_{t+h} = \mu_h + \theta_h^{LP} i_t + \gamma_h'(y_t, u_t)' + \sum_{\ell=1}^{p} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$ **VAR(p):** $\theta_h^{VAR} = [\Theta_h]_{y,i}/[\Theta_0]_{i,i}$



- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 1$: the VAR(1) is a smooth curve fitted to one autocovariance

C1: same shock, horizon by horizon

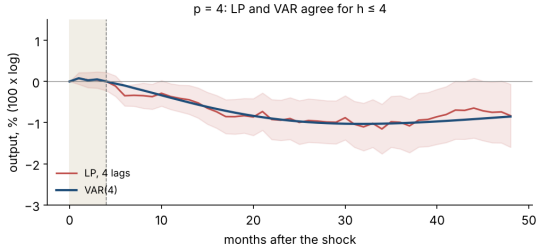
LP: $y_{t+h} = \mu_h + \theta_h^{LP} i_t + \gamma_h'(y_t, u_t)' + \sum_{\ell=1}^{p} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$ **VAR(p):** $\theta_h^{VAR} = [\Theta_h]_{y,i}/[\Theta_0]_{i,i}$



- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 2$: they agree up to $h = 2$; the VAR(2) is too flat afterwards (-0.75 vs -0.89 at 48)

C1: same shock, horizon by horizon

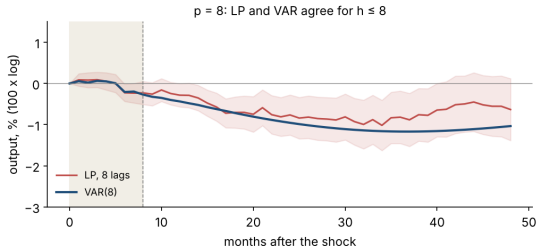
LP: $y_{t+h} = \mu_h + \theta_h^{LP} i_t + \gamma_h'(y_t, u_t)' + \sum_{\ell=1}^{p} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$ **VAR(p):** $\theta_h^{VAR} = [\Theta_h]_{y,i}/[\Theta_0]_{i,i}$



- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 4$: close up to $h = 4$, and here also beyond

C1: same shock, horizon by horizon

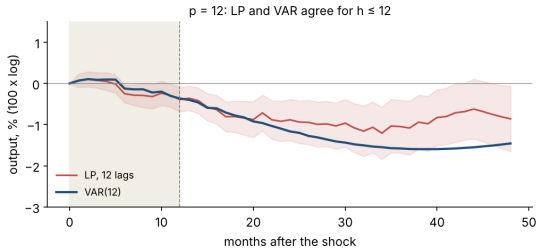
LP: $y_{t+h} = \mu_h + \theta_h^{LP} i_t + \gamma_h'(y_t, u_t)' + \sum_{\ell=1}^{p} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$ **VAR(p):** $\theta_h^{VAR} = [\Theta_h]_{y,i} / [\Theta_0]_{i,i}$



- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 8$: agree up to $h = 8$; beyond, the VAR(8) falls further (-1.17 vs -0.82 at 36)

C1: same shock, horizon by horizon

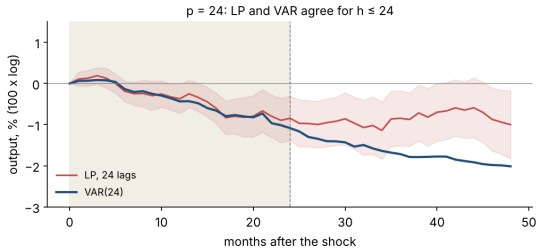
LP: $y_{t+h} = \mu_h + \theta_h^{LP} i_t + \gamma_h'(y_t, u_t)' + \sum_{\ell=1}^{p} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$ **VAR(p):** $\theta_h^{VAR} = [\Theta_h]_{y,i} / [\Theta_0]_{i,i}$



- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 12$: agree up to $h = 12$; at 36 months the VAR(12) says -1.58, the LP -1.05

C1: same shock, horizon by horizon

LP: $y_{t+h} = \mu_h + \theta_h^{LP} i_t + \gamma_h'(y_t, u_t)' + \sum_{\ell=1}^{p} \delta_{h,\ell}' x_{t-\ell} + \xi_{h,t+h}$ **VAR(p):** $\theta_h^{VAR} = [\Theta_h]_{y,i}/[\Theta_0]_{i,i}$



- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 24$: agree up to $h = 24$; beyond, the VAR(24) keeps falling to -2.0 at 48 months

What the sequence shows

What the sequence shows

Lags p	1	2	4	8	12	24
Max $\lvert LP - VAR \rvert$, $h \leq p$	0.16	0.04	0.01	0.06	0.15	0.24
Max $\lvert LP - VAR \rvert$, $h > p$	0.29	0.39	0.27	0.66	0.93	1.32
VAR(p) at $h = 48$	-0.92	-0.75	-0.85	-1.03	-1.46	-2.01
LP, p lags, at $h = 48$	-0.81	-0.89	-0.83	-0.63	-0.86	-1.00

What the sequence shows

Lags p	1	2	4	8	12	24
Max $\lvert LP - VAR \rvert$, $h \leq p$	0.16	0.04	0.01	0.06	0.15	0.24
Max $\lvert LP - VAR \rvert$, $h > p$	0.29	0.39	0.27	0.66	0.93	1.32
VAR(p) at $h = 48$	-0.92	-0.75	-0.85	-1.03	-1.46	-2.01
LP, p lags, at $h = 48$	-0.81	-0.89	-0.83	-0.63	-0.86	-1.00

- Up to $h = p$: same shock, same answer; exact in population, close in sample [Plagborg-Møller & Wolf (2021)]

What the sequence shows

Lags p	1	2	4	8	12	24
Max $\lvert LP - VAR \rvert$, $h \leq p$	0.16	0.04	0.01	0.06	0.15	0.24
Max $\lvert LP - VAR \rvert$, $h > p$	0.29	0.39	0.27	0.66	0.93	1.32
VAR(p) at $h = 48$	-0.92	-0.75	-0.85	-1.03	-1.46	-2.01
LP, p lags, at $h = 48$	-0.81	-0.89	-0.83	-0.63	-0.86	-1.00

- Up to $h = p$: same shock, same answer; exact in population, close in sample [Plagborg-Møller & Wolf (2021)]
- Beyond p: the VAR extrapolates its first p autocovariances; the LP keeps using the data

What the sequence shows

Lags p	1	2	4	8	12	24
Max $\lvert LP - VAR \rvert$, $h \leq p$	0.16	0.04	0.01	0.06	0.15	0.24
Max $\lvert LP - VAR \rvert$, $h > p$	0.29	0.39	0.27	0.66	0.93	1.32
VAR(p) at $h = 48$	-0.92	-0.75	-0.85	-1.03	-1.46	-2.01
LP, p lags, at $h = 48$	-0.81	-0.89	-0.83	-0.63	-0.86	-1.00

- Up to $h = p$: same shock, same answer; exact in population, close in sample [Plagborg-Møller & Wolf (2021)]
- Beyond p: the VAR extrapolates its first p autocovariances; the LP keeps using the data
- The VAR's long-run answer moves with p (from -0.75 to -2.0); the LP's hardly does

What the sequence shows

Lags p	1	2	4	8	12	24
Max $\lvert LP - VAR \rvert$, $h \leq p$	0.16	0.04	0.01	0.06	0.15	0.24
Max $\lvert LP - VAR \rvert$, $h > p$	0.29	0.39	0.27	0.66	0.93	1.32
VAR(p) at $h = 48$	-0.92	-0.75	-0.85	-1.03	-1.46	-2.01
LP, p lags, at $h = 48$	-0.81	-0.89	-0.83	-0.63	-0.86	-1.00

- Up to $h = p$: same shock, same answer; exact in population, close in sample [Plagborg-Møller & Wolf (2021)]
- Beyond p: the VAR extrapolates its first p autocovariances; the LP keeps using the data
- The VAR's long-run answer moves with p (from -0.75 to -2.0); the LP's hardly does

$\Rightarrow$ Which is closer to the truth is a bias-variance question [Plagborg-Møller (2026), lecture 2]

Internal-instrument VAR versus LP-IV

Internal-instrument VAR versus LP-IV

	VAR, z ordered first	**LP-IV**
Uses z for	Every horizon, via the VAR	Every horizon, directly
Needs invertibility	No	No
Identifying assumption	Lead-lag exogeneity of z	Same
Horizons beyond p	Extrapolated from the VAR	Estimated from the data
Precision	Higher	Lower

Internal-instrument VAR versus LP-IV

	VAR, z ordered first	**LP-IV**
Uses z for	Every horizon, via the VAR	Every horizon, directly
Needs invertibility	No	No
Identifying assumption	Lead-lag exogeneity of z	Same
Horizons beyond p	Extrapolated from the VAR	Estimated from the data
Precision	Higher	Lower

- **The bridge:** same estimand in population: LP-IV = recursive VAR with the instrument ordered first, normalized by impact on the policy variable [Plagborg-Møller & Wolf (2021)]

Internal-instrument VAR versus LP-IV

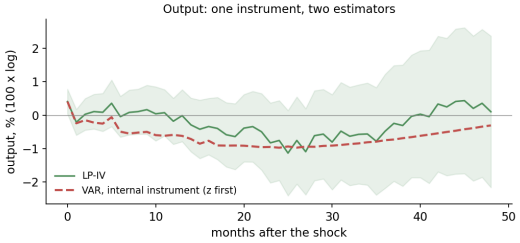
	VAR, z ordered first	**LP-IV**
Uses z for	Every horizon, via the VAR	Every horizon, directly
Needs invertibility	No	No
Identifying assumption	Lead-lag exogeneity of z	Same
Horizons beyond p	Extrapolated from the VAR	Estimated from the data
Precision	Higher	Lower

- **The bridge:** same estimand in population: LP-IV = recursive VAR with the instrument ordered first, normalized by impact on the policy variable [Plagborg-Møller & Wolf (2021)]

$\Rightarrow$ Identification is the same; they differ only in estimation: next, bias vs variance

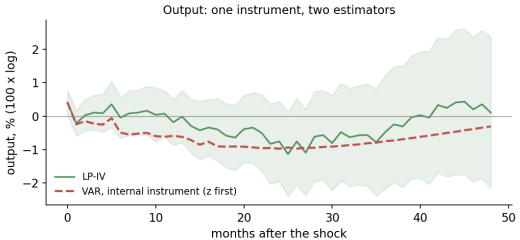
C2: one instrument, two estimators

VAR with z first (V4) vs. LP-IV: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_\ell \delta'_{h,\ell} w_{t-\ell} + \xi_{h,t+h}$ (L3)



C2: one instrument, two estimators

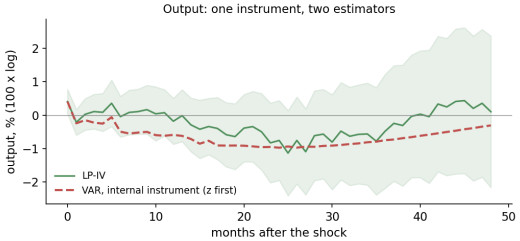
VAR with z first (V4) vs. LP-IV: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_\ell \delta'_{h,\ell} w_{t-\ell} + \xi_{h,t+h}$ (L3)



- Output at 12 / 24 / 36 months: VAR with z first -0.59 / -0.98 / -0.75; LP-IV -0.18 / -0.76 / -0.48

C2: one instrument, two estimators

VAR with z first (V4) vs. LP-IV: $y_{t+h} = \mu_h + \theta_h \hat{i}_t + \sum_\ell \delta'_{h,\ell} w_{t-\ell} + \xi_{h,t+h}$ (L3)



- Output at 12 / 24 / 36 months: VAR with z first -0.59 / -0.98 / -0.75; LP-IV -0.18 / -0.76 / -0.48

$\Rightarrow$ Both put output $\sim$0.8–1% lower at 2 years; the VAR is smoother, LP-IV noisier: the trade-off of the next slides

Why VARs and LPs differ in finite samples

Why VARs and LPs differ in finite samples

- A VAR is a good approximation, not the literal truth: small dynamics are always left out (lags, variables, measurement error)

Why VARs and LPs differ in finite samples

- A VAR is a good approximation, not the literal truth: small dynamics are always left out (lags, variables, measurement error)
- **Why the LP is robust to omitting lags:** θ_h only needs the date-t shock to be clean. Omitted dynamics end up in the error term: noisier, but not biased, since the shock is uncorrelated with them

Why VARs and LPs differ in finite samples

- A VAR is a good approximation, not the literal truth: small dynamics are always left out (lags, variables, measurement error)
- **Why the LP is robust to omitting lags:** θ_h only needs the date-t shock to be clean. Omitted dynamics end up in the error term: noisier, but not biased, since the shock is uncorrelated with them
- **Why the VAR is not:** it estimates one-month dynamics $\hat{A}$ and builds every horizon from them (Θ_{12} uses $\hat{A}^{12}$). A tiny error in $\hat{A}$ compounds: $0.85^{12} = 0.14$, $0.83^{12} = 0.11$ (25% off)

Why VARs and LPs differ in finite samples

- A VAR is a good approximation, not the literal truth: small dynamics are always left out (lags, variables, measurement error)
- **Why the LP is robust to omitting lags:** θ_h only needs the date-t shock to be clean. Omitted dynamics end up in the error term: noisier, but not biased, since the shock is uncorrelated with them
- **Why the VAR is not:** it estimates one-month dynamics $\hat{A}$ and builds every horizon from them (Θ_{12} uses $\hat{A}^{12}$). A tiny error in $\hat{A}$ compounds: $0.85^{12} = 0.14$, $0.83^{12} = 0.11$ (25% off)
- **Formally:** truth = VAR(p_0) + small MA terms; with $p \geq p_0$ lags, in large samples [Li, Plagborg-Møller & Wolf (2024)]

$$\hat{\theta}_h^{VAR} \dot{\sim} N\big(\theta_h + b_h(p),\ \tau^2_{h,VAR}(p)\big), \qquad \hat{\theta}_h^{LP} \dot{\sim} N\big(\theta_h,\ \tau^2_{h,LP}\big), \qquad \tau^2_{h,VAR}(p) \leq \tau^2_{h,LP}$$

Why VARs and LPs differ in finite samples

- A VAR is a good approximation, not the literal truth: small dynamics are always left out (lags, variables, measurement error)
- **Why the LP is robust to omitting lags:** θ_h only needs the date-t shock to be clean. Omitted dynamics end up in the error term: noisier, but not biased, since the shock is uncorrelated with them
- **Why the VAR is not:** it estimates one-month dynamics $\hat{A}$ and builds every horizon from them (Θ_{12} uses $\hat{A}^{12}$). A tiny error in $\hat{A}$ compounds: $0.85^{12} = 0.14$, $0.83^{12} = 0.11$ (25% off)
- **Formally:** truth = VAR(p_0) + small MA terms; with $p \geq p_0$ lags, in large samples [Li, Plagborg-Møller & Wolf (2024)]

$$\hat{\theta}_h^{VAR} \dot{\sim} N\big(\theta_h + b_h(p),\ \tau^2_{h,VAR}(p)\big), \qquad \hat{\theta}_h^{LP} \dot{\sim} N\big(\theta_h,\ \tau^2_{h,LP}\big), \qquad \tau^2_{h,VAR}(p) \leq \tau^2_{h,LP}$$

$\Rightarrow$ Both need the important lags; LP is robust to omitting moderately important ones, the VAR is not

The bias-variance trade-off

The bias-variance trade-off

- **VAR:** every horizon from the same few coefficients, estimated on all the data $\rightarrow$ smooth and precise (low variance); if the structure is slightly wrong, every horizon inherits it (bias)

The bias-variance trade-off

- **VAR:** every horizon from the same few coefficients, estimated on all the data $\rightarrow$ smooth and precise (low variance); if the structure is slightly wrong, every horizon inherits it (bias)
- **LP:** one regression per h, nothing imposed across horizons $\rightarrow$ unbiased, but serially correlated errors and fewer usable observations (high variance)

The bias-variance trade-off

- **VAR:** every horizon from the same few coefficients, estimated on all the data $\rightarrow$ smooth and precise (low variance); if the structure is slightly wrong, every horizon inherits it (bias)
- **LP:** one regression per h, nothing imposed across horizons $\rightarrow$ unbiased, but serially correlated errors and fewer usable observations (high variance)
- **Picture:** a smooth curve vs connecting the dots. Up to $h \leq p - p_0$ the two are identical

The bias-variance trade-off

- **VAR:** every horizon from the same few coefficients, estimated on all the data $\rightarrow$ smooth and precise (low variance); if the structure is slightly wrong, every horizon inherits it (bias)
- **LP:** one regression per h, nothing imposed across horizons $\rightarrow$ unbiased, but serially correlated errors and fewer usable observations (high variance)
- **Picture:** a smooth curve vs connecting the dots. Up to $h \leq p - p_0$ the two are identical
- **The uncomfortable part:** misspecification too small to detect can make VAR bias as large as its SE

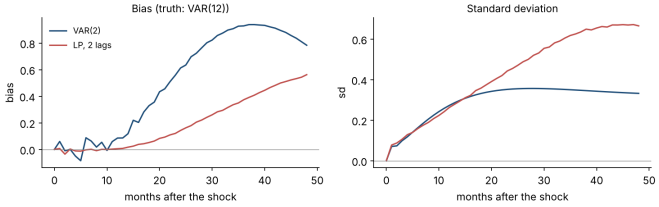
The bias-variance trade-off

- **VAR:** every horizon from the same few coefficients, estimated on all the data → smooth and precise (low variance); if the structure is slightly wrong, every horizon inherits it (bias)
- **LP:** one regression per h, nothing imposed across horizons → unbiased, but serially correlated errors and fewer usable observations (high variance)
- **Picture:** a smooth curve vs connecting the dots. Up to $h \leq p - p_0$ the two are identical
- **The uncomfortable part:** misspecification too small to detect can make VAR bias as large as its SE
- **Formally** ("no free lunch"), M = share of MA-residual variance from lagged shocks: [Li, Plagborg-Møller & Wolf (2024)]

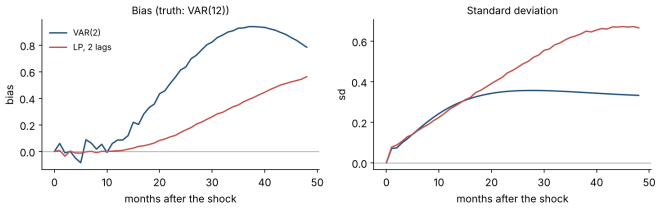
$$|b_h(p)| \leq \sqrt{T \times M \times \left(\tau^2_{h,LP} - \tau^2_{h,VAR}(p)\right)}, \qquad \text{and the bound is attained}$$

$T = 100$, $M = 1\%$, VAR variance half of LP's → worst-case VAR bias = 1.7 standard errors

C3: the trade-off on our data

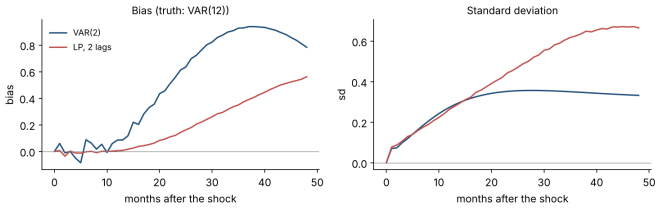


C3: the trade-off on our data



- Truth: the running example's VAR(12); 500 samples of $T = 356$, output response estimated with 2 lags

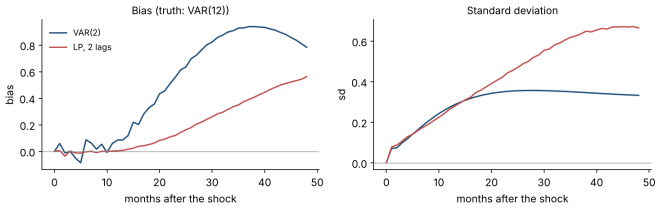
C3: the trade-off on our data



- Truth: the running example's VAR(12); 500 samples of $T = 356$, output response estimated with 2 lags

$h = 13$–48	Mean $\lvert\text{bias}\rvert$	Mean sd
VAR(2)	0.68	0.34
LP(2)	0.27	0.53

C3: the trade-off on our data



- Truth: the running example's VAR(12); 500 samples of $T = 356$, output response estimated with 2 lags

$h = 13$–48	Mean $\lvert\text{bias}\rvert$	Mean sd
VAR(2)	0.68	0.34
LP(2)	0.27	0.53

$\Rightarrow$ VAR bias is twice its sd; LP bias is smaller (not zero: 2 lags do not pin down the true shock) but noisier

Uncertainty: bias is costly for confidence bands

Uncertainty: bias is costly for confidence bands

- We want 90% bands that cover for every plausible DGP, not only on average

Uncertainty: bias is costly for confidence bands

- We want 90% bands that cover for every plausible DGP, not only on average
- A biased VAR band has the right width but is off-centre: with bias $\approx 1.7 \times SE$, a nominal 90% band covers far less

Uncertainty: bias is costly for confidence bands

- We want 90% bands that cover for every plausible DGP, not only on average
- A biased VAR band has the right width but is off-centre: with bias $\approx 1.7 \times SE$, a nominal 90% band covers far less
- **Simulation evidence** (200 stationary + 200 non-stationary DGPs, $T = 240$):
  - MSE: (B)VAR with few lags beats LP on average; shrinkage helps both
  - Coverage $\geq 80\%$ (target 90%): LP in most DGPs; short VARs fail at medium and long horizons

Uncertainty: bias is costly for confidence bands

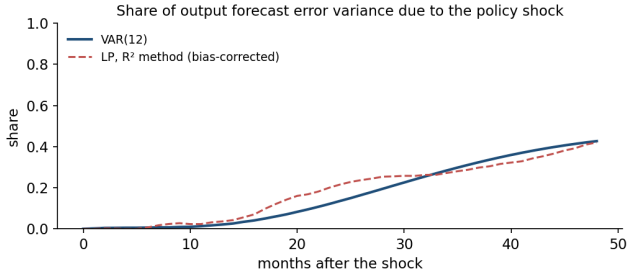
- We want 90% bands that cover for every plausible DGP, not only on average
- A biased VAR band has the right width but is off-centre: with bias $\approx 1.7 \times SE$, a nominal 90% band covers far less
- **Simulation evidence** (200 stationary + 200 non-stationary DGPs, $T = 240$):
  - MSE: (B)VAR with few lags beats LP on average; shrinkage helps both
  - Coverage $\geq$ 80% (target 90%): LP in most DGPs; short VARs fail at medium and long horizons

$\Rightarrow$ Honest bands need an LP, or a VAR with many more lags than usual

[Montiel Olea, Plagborg-Møller, Qian & Wolf (2025)]

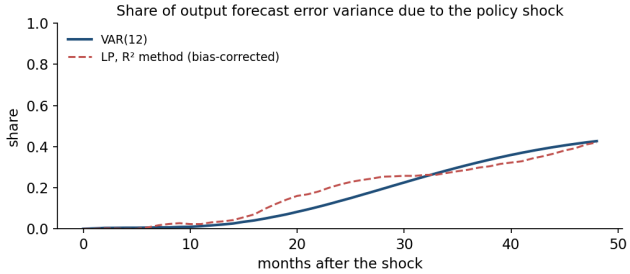
C4: variance decompositions side by side

VAR: $\dfrac{\sum_{\ell<h}[\Theta_\ell]^2_{y,i}}{\sum_{\ell<h}\sum_m[\Theta_\ell]^2_{y,m}}$ (V5) **LP:** R^2 of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$ (L6)



C4: variance decompositions side by side

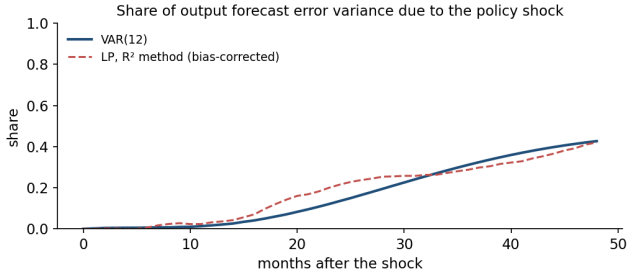
VAR: $\dfrac{\sum_{\ell<h}[\Theta_\ell]^2_{y,i}}{\sum_{\ell<h}\sum_m[\Theta_\ell]^2_{y,m}}$ (V5) **LP:** R^2 of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$ (L6)



- 12 / 24 / 48 months: VAR 1% / 12% / 42%; LP (bias-corrected) 2% / 20% / 41%

C4: variance decompositions side by side

VAR: $\dfrac{\sum_{\ell<h}[\Theta_\ell]^2_{y,i}}{\sum_{\ell<h}\sum_m[\Theta_\ell]^2_{y,m}}$ (V5) **LP:** R^2 of $\hat{f}_{h,t+h}$ on $\tilde{x}_t,\ldots,\tilde{x}_{t+h}$ (L6)



- 12 / 24 / 48 months: VAR 1% / 12% / 42%; LP (bias-corrected) 2% / 20% / 41%

$\Rightarrow$ Same shock, same long-run answer; the LP finds policy more important earlier

Long horizons and persistent data

Long horizons and persistent data

- **VAR:** response is a high power of $\hat{A} \to$ nonlinear; normal approximation breaks down at long h or near unit roots

Long horizons and persistent data

- **VAR:** response is a high power of $\hat{A} \to$ nonlinear; normal approximation breaks down at long h or near unit roots
- **LP:** linear regression at every h; with lag augmentation the shock is stationary $\to$ inference holds at moderate horizons, even with unit roots

Long horizons and persistent data

- **VAR:** response is a high power of $\hat{A} \to$ nonlinear; normal approximation breaks down at long h or near unit roots
- **LP:** linear regression at every h; with lag augmentation the shock is stationary $\to$ inference holds at moderate horizons, even with unit roots
- **But:** LP also has small-sample bias with persistent data ($T \approx 100$): it understates persistence [Herbst & Johannsen (2024)]

Fixes for persistent data

Fixes for persistent data

- **Long differences** with an observed (serially uncorrelated) shock [Piger & Stockwell (2025)]

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{long difference}} = \mu_h + \theta_h \varepsilon_t + \underbrace{\delta'_{h,1}\Delta w_{t-1} + \cdots + \delta'_{h,p}\Delta w_{t-p}}_{\text{lagged first differences}} + \xi_{h,t+h}$$

  - Less persistent regressors $\rightarrow$ smaller bias; may cost efficiency

Fixes for persistent data

- **Long differences** with an observed (serially uncorrelated) shock [Piger & Stockwell (2025)]

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{long difference}} = \mu_h + \theta_h \varepsilon_t + \underbrace{\delta'_{h,1}\Delta w_{t-1} + \cdots + \delta'_{h,p}\Delta w_{t-p}}_{\text{lagged first differences}} + \xi_{h,t+h}$$

  - Less persistent regressors $\rightarrow$ smaller bias; may cost efficiency

- **Analytical bias correction:** works for general LPs; costs variance in small samples [Herbst & Johannsen (2024)]

Fixes for persistent data

- **Long differences** with an observed (serially uncorrelated) shock [Piger & Stockwell (2025)]

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{long difference}} = \mu_h + \theta_h \varepsilon_t + \underbrace{\delta'_{h,1}\Delta w_{t-1} + \cdots + \delta'_{h,p}\Delta w_{t-p}}_{\text{lagged first differences}} + \xi_{h,t+h}$$

  - Less persistent regressors $\rightarrow$ smaller bias; may cost efficiency
- **Analytical bias correction:** works for general LPs; costs variance in small samples [Herbst & Johannsen (2024)]
- **VAR-based bootstrap** for LP bands [Montiel Olea et al. (2025)]
  1. Estimate an auxiliary VAR; simulate fake data from it
  2. Apply the LP to each fake sample; repeat; take percentiles (can also bias-correct)

Fixes for persistent data

- **Long differences** with an observed (serially uncorrelated) shock [Piger & Stockwell (2025)]

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{long difference}} = \mu_h + \theta_h \varepsilon_t + \underbrace{\delta'_{h,1}\Delta w_{t-1} + \cdots + \delta'_{h,p}\Delta w_{t-p}}_{\text{lagged first differences}} + \xi_{h,t+h}$$

  - Less persistent regressors $\rightarrow$ smaller bias; may cost efficiency
- **Analytical bias correction:** works for general LPs; costs variance in small samples [Herbst & Johannsen (2024)]
- **VAR-based bootstrap** for LP bands [Montiel Olea et al. (2025)]
  1. Estimate an auxiliary VAR; simulate fake data from it
  2. Apply the LP to each fake sample; repeat; take percentiles (can also bias-correct)

$\Rightarrow$ Matters mostly at long horizons with persistent data: run it as a robustness check

[Plagborg-Møller (2026), lecture 3]

So which should you use?

So which should you use?

- **If you care about the point estimate (MSE):**
  - VAR, ideally with shrinkage (BVAR), wins in the typical simulated DGP
  - Shrunk LPs help too

So which should you use?

- **If you care about the point estimate (MSE):**
  - VAR, ideally with shrinkage (BVAR), wins in the typical simulated DGP
  - Shrunk LPs help too
- **If you care about honest uncertainty (coverage):**
  - LP with lag augmentation, or a VAR with many more lags than usual
  - Short VARs have bands that under-cover at medium and long horizons

So which should you use?

- **If you care about the point estimate (MSE):**
  - VAR, ideally with shrinkage (BVAR), wins in the typical simulated DGP
  - Shrunk LPs help too
- **If you care about honest uncertainty (coverage):**
  - LP with lag augmentation, or a VAR with many more lags than usual
  - Short VARs have bands that under-cover at medium and long horizons

$\Rightarrow$ Depends on your loss function. In a thesis, referees mostly ask whether your bands are honest

[Li, Plagborg-Møller & Wolf (2024), Montiel Olea et al. (2025)]

Recommendations for your own work

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can
2. **Main specification:** lag-augmented LP with robust SEs, or VAR with generous lags; show the other as robustness

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can
2. **Main specification:** lag-augmented LP with robust SEs, or VAR with generous lags; show the other as robustness
3. **Instruments:** LP-IV or internal-instrument VAR; report first-stage F

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can
2. **Main specification:** lag-augmented LP with robust SEs, or VAR with generous lags; show the other as robustness
3. **Instruments:** LP-IV or internal-instrument VAR; report first-stage F
4. **Diagnostics:** plot the implicit shock, report its scale, check it is unpredictable

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can
2. **Main specification:** lag-augmented LP with robust SEs, or VAR with generous lags; show the other as robustness
3. **Instruments:** LP-IV or internal-instrument VAR; report first-stage F
4. **Diagnostics:** plot the implicit shock, report its scale, check it is unpredictable
5. **Bands:** 68% and 90% pointwise; sup-t band for claims across horizons

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can
2. **Main specification:** lag-augmented LP with robust SEs, or VAR with generous lags; show the other as robustness
3. **Instruments:** LP-IV or internal-instrument VAR; report first-stage F
4. **Diagnostics:** plot the implicit shock, report its scale, check it is unpredictable
5. **Bands:** 68% and 90% pointwise; sup-t band for claims across horizons
6. **Persistent data, long horizons:** bootstrap, or long differences

Recommendations for your own work

1. **Decide identification first.** Measure a shock or proxy directly if you can
2. **Main specification:** lag-augmented LP with robust SEs, or VAR with generous lags; show the other as robustness
3. **Instruments:** LP-IV or internal-instrument VAR; report first-stage F
4. **Diagnostics:** plot the implicit shock, report its scale, check it is unpredictable
5. **Bands:** 68% and 90% pointwise; sup-t band for claims across horizons
6. **Persistent data, long horizons:** bootstrap, or long differences
7. **Importance:** report a variance decomposition next to the responses

A workflow, start to finish

A workflow, start to finish

Step	What you do	Example
1. Data	Plot everything; levels fine with enough lags [Sims, Stock & Watson (1990)]	L1
2. Lags	AIC on an auxiliary VAR; lean long	L2, L4, V3
3. Shock	Ordering, instrument or signs: state the assumption in one sentence	L3, V1, V2, V4
4. Estimate	SVAR and LP with the same shock and the same p	C1, C2
5. Inference	Bootstrap for the VAR, robust SEs for the lag-augmented LP	L5, V1
6. Importance	Variance decomposition	L6, V5, C4
7. Robustness	More lags, other ordering/instrument, LP vs VAR	V2, V3, C1–C3

Checkpoint

Checkpoint

- **Key take-aways**:

  $\Rightarrow$ Identification is economics; estimation is statistics. Keep them apart

Checkpoint

- **Key take-aways**:

  $\Rightarrow$ Identification is economics; estimation is statistics. Keep them apart

  $\Rightarrow$ SVAR picks Θ_0; LP picks controls – both define the same shock

Checkpoint

- **Key take-aways**:
  - $\Rightarrow$ Identification is economics; estimation is statistics. Keep them apart
  - $\Rightarrow$ SVAR picks Θ_0; LP picks controls – both define the same shock
  - $\Rightarrow$ LP and VAR agree up to the lag length; beyond it the VAR extrapolates

Checkpoint

- **Key take-aways**:
  - $\Rightarrow$ Identification is economics; estimation is statistics. Keep them apart
  - $\Rightarrow$ SVAR picks Θ_0; LP picks controls – both define the same shock
  - $\Rightarrow$ LP and VAR agree up to the lag length; beyond it the VAR extrapolates
  - $\Rightarrow$ LP: low bias, high variance, honest bands. Short VAR: precise, but possibly biased in ways the bands do not show

Checkpoint

- **Key take-aways**:
  - $\Rightarrow$ Identification is economics; estimation is statistics. Keep them apart
  - $\Rightarrow$ SVAR picks Θ_0; LP picks controls – both define the same shock
  - $\Rightarrow$ LP and VAR agree up to the lag length; beyond it the VAR extrapolates
  - $\Rightarrow$ LP: low bias, high variance, honest bands. Short VAR: precise, but possibly biased in ways the bands do not show
  - $\Rightarrow$ Lag augmentation, the shock's scale, a variance decomposition, joint bands for joint claims

Checkpoint

- **Key take-aways**:
  - $\Rightarrow$ Identification is economics; estimation is statistics. Keep them apart
  - $\Rightarrow$ SVAR picks Θ_0; LP picks controls – both define the same shock
  - $\Rightarrow$ LP and VAR agree up to the lag length; beyond it the VAR extrapolates
  - $\Rightarrow$ LP: low bias, high variance, honest bands. Short VAR: precise, but possibly biased in ways the bands do not show
  - $\Rightarrow$ Lag augmentation, the shock's scale, a variance decomposition, joint bands for joint claims

- Next: implementation: the same steps in code

Implementation

Lab: the running example in code

Lab: the running example in code

- **Two scripts, same numbers** (`Week1_Lab`):
  - `week1_simple.py`: one cell per step, every LP a regression formula (statsmodels, linearmodels)
  - `week1_fast.py`: the same cells in fast numpy (all 500 bootstrap samples at once); makes the slide figures

Lab: the running example in code

- **Two scripts, same numbers** (`Week1_Lab`):
  - `week1_simple.py`: one cell per step, every LP a regression formula (statsmodels, linearmodels)
  - `week1_fast.py`: the same cells in fast numpy (all 500 bootstrap samples at once); makes the slide figures
- **In class** ($\sim$25 minutes): run `week1_simple.py` cell by cell: L1–L6, V1–V5, C1–C4
  - Change one thing per step: lags, ordering, instrument, standard errors

LPs in code: a loop over h

```
def ols(Y, X):                                    # drops rows with NaN
    keep = ~np.isnan(np.column_stack([Y, X])).any(axis=1)
    Y, X = Y[keep], X[keep]
    n, k = X.shape
    XXi = np.linalg.inv(X.T @ X)
    b = XXi @ X.T @ Y
    e = Y - X @ b
    V = XXi @ ((X * e[:, None] ** 2).T @ X) @ XXi * n / (n - k)   # HC1
    return b, np.sqrt(np.diag(V)), e, X

# L2(c): output on the rate, controls y_t, u_t and 12 lags of y, u, i
W = np.column_stack([one, y, u, lagged([y, u, i], 12)])
out = np.zeros((H + 1, 2))
for h in range(H + 1):
    b, se, _, _ = ols(lead(y, h), np.column_stack([i, W]))
    out[h] = b[0], se[0]                          # theta_h and its SE
```

$\Rightarrow$ Only the left-hand side shifts across horizons: that is the whole appeal

The VAR in code: estimate once, iterate forward

```
def var_ols(data, p):                          # each equation: OLS on p lags of all
    n = len(data)
    Y = data[p:]
    X = np.column_stack([np.ones(n - p)] + [data[p - l:n - l] for l in range(1, p + 1)])
    B = np.linalg.inv(X.T @ X) @ X.T @ Y
    U = Y - X @ B
    A = [B[1 + (l - 1) * k: 1 + l * k].T for l in range(1, p + 1)]   # A_1..A_p
    return A, U, U.T @ U / (n - p - X.shape[1]), B[0]

def var_irf(A, impact, H):                     # Theta_h = A_1 Theta_{h-1} + ... + A_p Theta_{h-p}
    irf = np.zeros((H + 1, len(impact)))
    irf[0] = impact
    for h in range(1, H + 1):
        for l in range(1, min(h, len(A)) + 1):
            irf[h] += A[l - 1] @ irf[h - l]
    return irf

A, U, Sigma, c = var_ols(np.column_stack([y, u, i]), 12)
P = np.linalg.cholesky(Sigma)                  # recursive: order (y, u, i)
var12 = var_irf(A, P[:, 2] / P[2, 2], H)       # V1: +1pp on the rate
```

Three ways to use the narrative shock

```
# 1. SVAR-IV: VAR residuals on z, then iterate the VAR
zt = z[12:] - z[12:].mean()
slopes = U.T @ zt / (zt @ zt)                    # Cov(u_t, z_t)
svar_iv = var_irf(A, slopes / slopes[2], H)      # +1pp on the rate

# 2. LP-IV (L3): 2SLS of y_{t+h} on i_t, instrument z_t, 12 lags of y, u, i, z
W = np.column_stack([one, lagged([y, u, i, z], 12)])
iv12 = np.array([iv(lead(y, h), i, z, W)[:2] for h in range(H + 1)])

# 3. Internal instrument: recursive VAR in (z, y, u, i), z ordered first
Az, Uz, Sz, _ = var_ols(np.column_stack([z, y, u, i]), 12)
Pz = np.linalg.cholesky(Sz)
svar_int = var_irf(Az, Pz[:, 0] / Pz[3, 0], H)[:, 1:]   # normalise on the rate
```

$\Rightarrow$ 2 and 3 estimate the same object and do not need invertibility; 1 does

Reading group

Gertler & Karadi (2015), AEJ: Macro

Gertler & Karadi (2015), AEJ: Macro

- **Question:** does monetary policy move credit costs by more than expected short rates, and what happens to activity?

Gertler & Karadi (2015), AEJ: Macro

- **Question:** does monetary policy move credit costs by more than expected short rates, and what happens to activity?
- **Contribution:** proxy SVAR with high-frequency futures surprises
  $\rightarrow$ no timing restrictions on financial variables

Gertler & Karadi (2015), AEJ: Macro

- **Question:** does monetary policy move credit costs by more than expected short rates, and what happens to activity?
- **Contribution:** proxy SVAR with high-frequency futures surprises
  $\rightarrow$ no timing restrictions on financial variables
- **Identification:** monthly VAR (1-year rate, IP, CPI, excess bond premium); instrument = 3-month-ahead fed funds futures surprise around FOMC announcements

Gertler & Karadi (2015), AEJ: Macro

- **Question:** does monetary policy move credit costs by more than expected short rates, and what happens to activity?
- **Contribution:** proxy SVAR with high-frequency futures surprises
  $\rightarrow$ no timing restrictions on financial variables
- **Identification:** monthly VAR (1-year rate, IP, CPI, excess bond premium); instrument = 3-month-ahead fed funds futures surprise around FOMC announcements
- **Main result:** tightening raises excess bond premium and term premia; output falls, prices fall modestly $\rightarrow$ evidence for a credit channel

Gertler & Karadi (2015), AEJ: Macro

- **Question:** does monetary policy move credit costs by more than expected short rates, and what happens to activity?
- **Contribution:** proxy SVAR with high-frequency futures surprises
  $\rightarrow$ no timing restrictions on financial variables
- **Identification:** monthly VAR (1-year rate, IP, CPI, excess bond premium); instrument = 3-month-ahead fed funds futures surprise around FOMC announcements
- **Main result:** tightening raises excess bond premium and term premia; output falls, prices fall modestly $\rightarrow$ evidence for a credit channel
- **Our exercise** (`exercise_PT_student.py`): LPs for Portugal on exogenous ECB high-frequency shocks [Jarociński & Karadi (2020)], split by the stock market: rates up, stocks down = MP; rates up, stocks up = CBI (information)

Gertler & Karadi (2015), AEJ: Macro

- **Question:** does monetary policy move credit costs by more than expected short rates, and what happens to activity?
- **Contribution:** proxy SVAR with high-frequency futures surprises
  $\rightarrow$ no timing restrictions on financial variables
- **Identification:** monthly VAR (1-year rate, IP, CPI, excess bond premium); instrument = 3-month-ahead fed funds futures surprise around FOMC announcements
- **Main result:** tightening raises excess bond premium and term premia; output falls, prices fall modestly $\rightarrow$ evidence for a credit channel
- **Our exercise** (`exercise_PT_student.py`): LPs for Portugal on exogenous ECB high-frequency shocks [Jarociński & Karadi (2020)], split by the stock market: rates up, stocks down = MP; rates up, stocks up = CBI (information)

$\Rightarrow$ Template for your presentations: question, contribution, identification, main result, weaknesses, one extension

Reading group and next week

Reading group and next week

Rotation

Week	Presents	Discusses
2	[Student A]	[Student B]
3	[Student B]	[Student C]
4	[Student C]	[Student A]
5	Project pitches	All